Lecture 9: Stress Transformation

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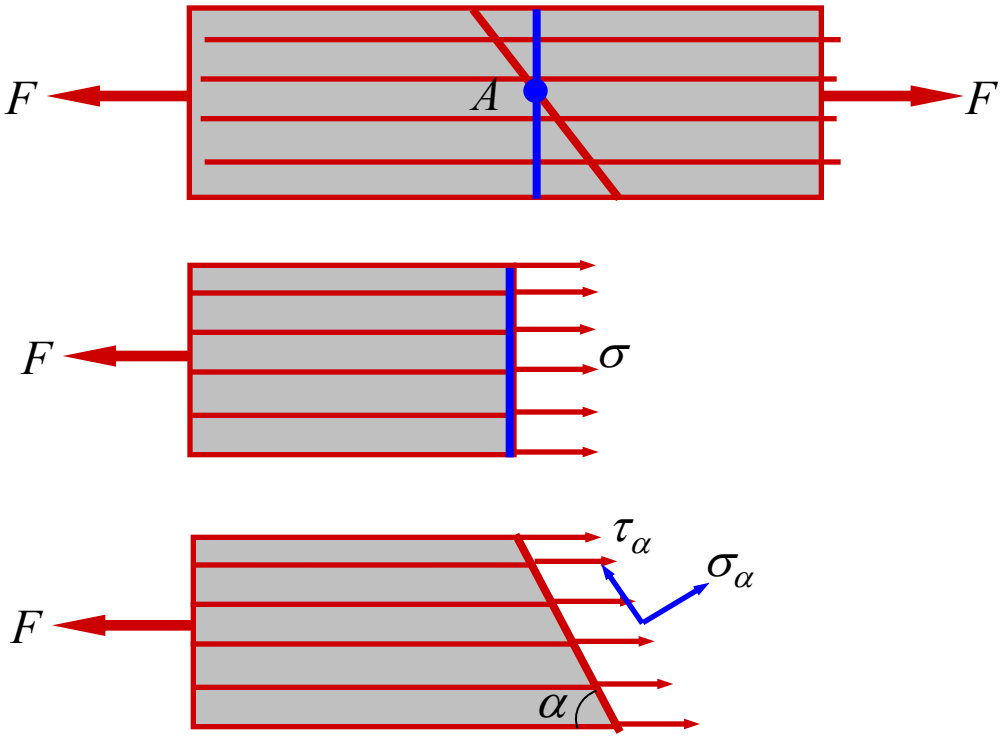
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The Stress State of a Point

- The stress state at a point
  - is defined as the collection of the stress distributions on all planes passing through the point.
  - can be analyzed via the stress distributions acting on a differential cube, arbitrarily oriented with reference coordinates
- Method of differential cubes with:
  - infinitesimal side length
  - uniformly distributed stress on each of the six surfaces
  - equal and opposite stress on parallel sides

Stresses on Oblique Planes

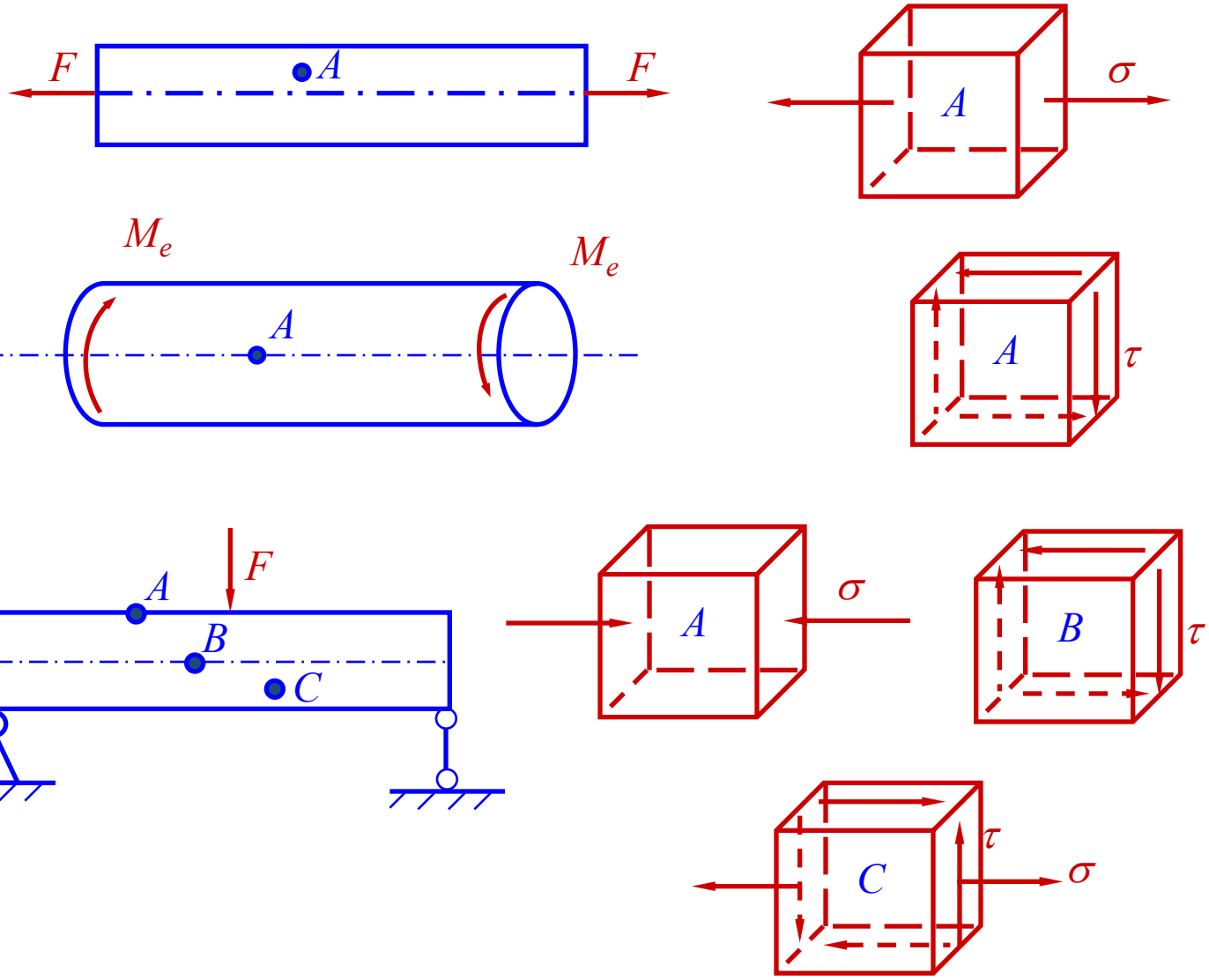


- The stress distribution at a point depends on the plane orientation along which the point is examined. Consider an oblique plane with angle α (bottom figure):

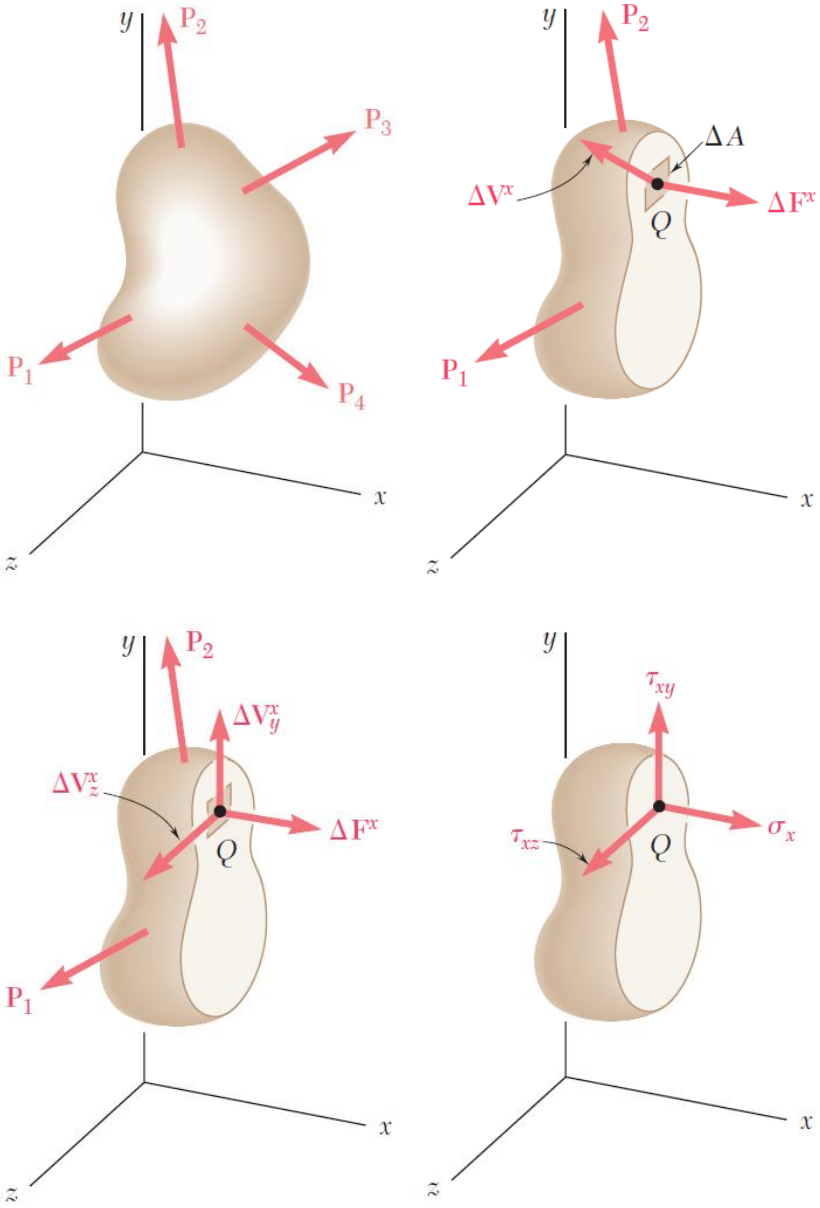
$$\sigma_\alpha = \sigma \cos^2 \alpha$$

$$\tau_\alpha = -\frac{\sigma}{2} \sin 2\alpha$$

Examples of the Stress State of a Point



A Deformable Body Under General Loadings



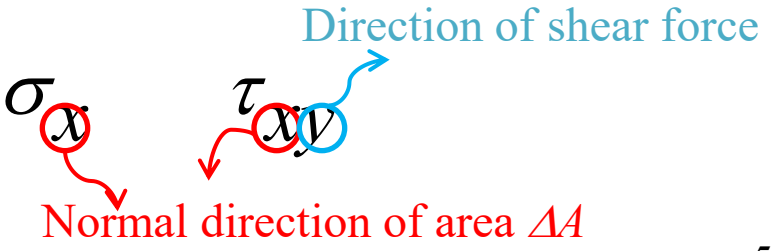
- A deformable body subjected to a general combination of loads is cut into two segments by a plane x passing through a point of interest Q
- Stress components may be defined as,

Normal Stress: $$\sigma_x = \lim_{\Delta A \to 0} \frac{\Delta F^x}{\Delta A}$$

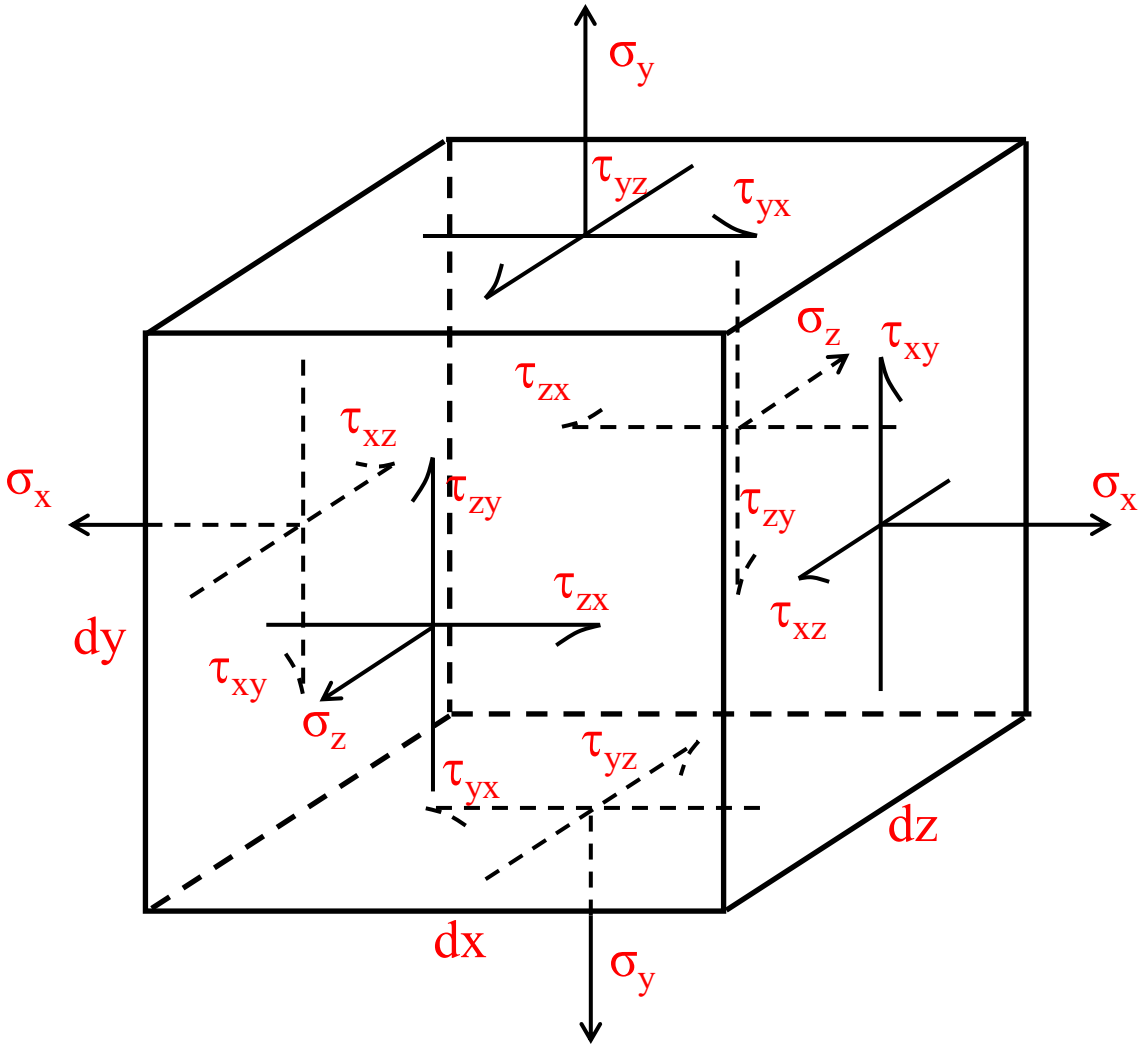
Shear Stress: $$\tau_{xy} = \lim_{\Delta A \to 0} \frac{\Delta V_y^x}{\Delta A}$$

$$\tau_{xz} = \lim_{\Delta A \to 0} \frac{\Delta V_z^x}{\Delta A}$$

Unit: $[\frac{\mathrm{N}}{\mathrm{m}^2}]$ or Pa

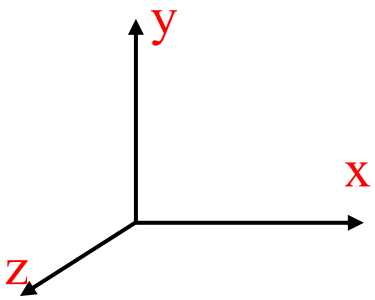


General Stress State of a Point



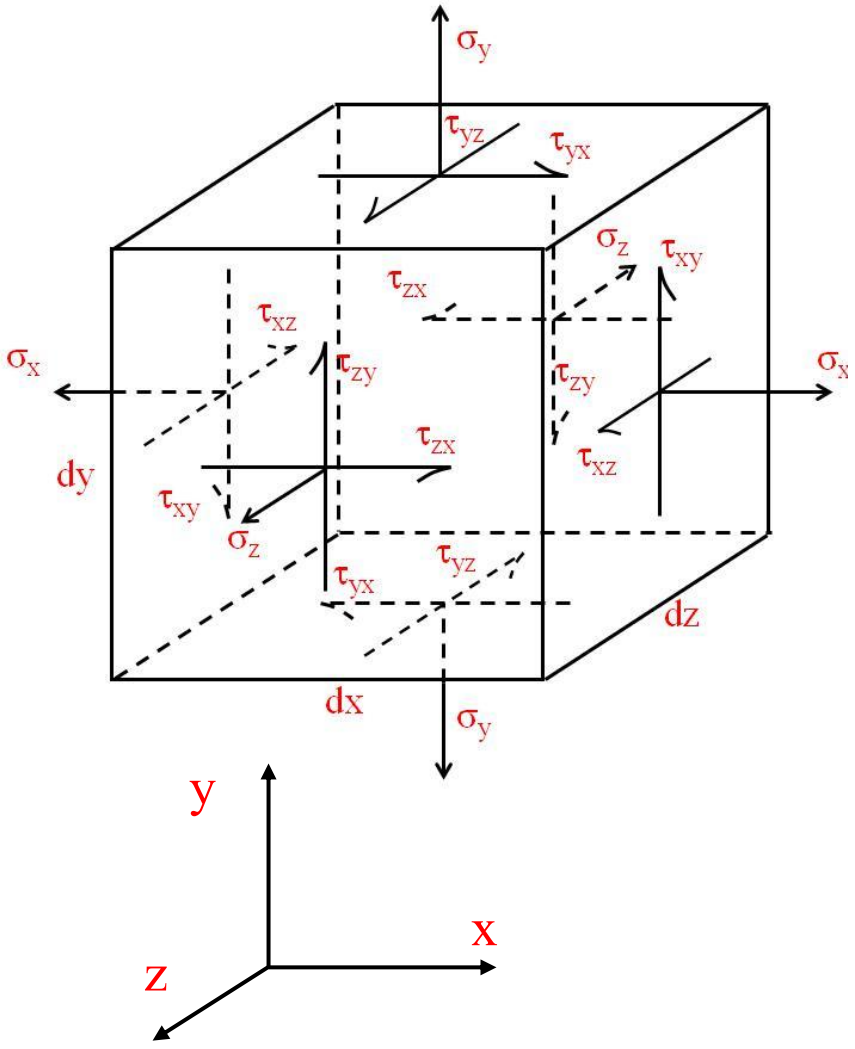
von Karman's Notation:
Symmetric

$$\left[\sigma_{ij}\right] = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$



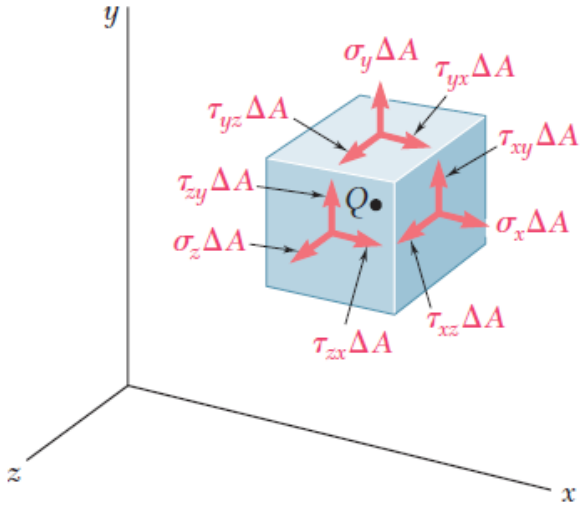
All stress components shown in the above figure are positive.

Sign Convention of Stress



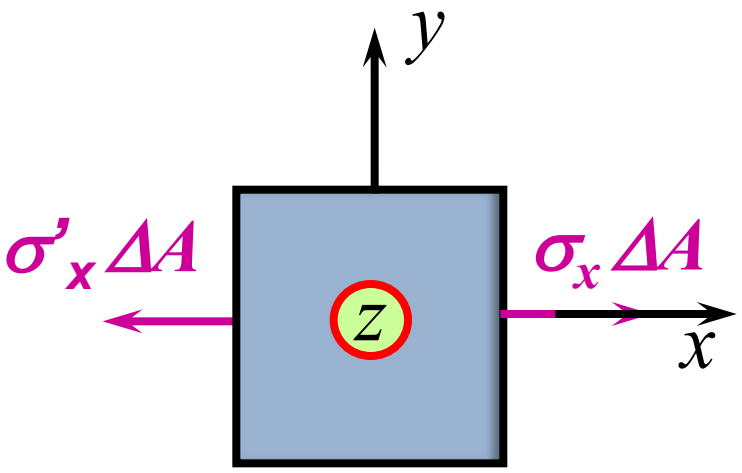
- **Normal stress:**
  positive along normal vector of cubic surface (or tensile direction);
- **Shear stress:**
  A shear stress acting on a positive face of an element is positive if it acts in the positive direction of one of the coordinate axes and negative if it acts in the negative direction of an axis. A shear stress acting on a negative face of an element is positive if it acts in the negative direction of an axis and negative if it acts in a positive direction.

Symmetry of Stress Components



- The combination of forces generated by the stresses must satisfy the equilibrium conditions:

$$\sum F_x = \sum F_y = \sum F_z = 0$$
$$\sum M_x = \sum M_y = \sum M_z = 0$$

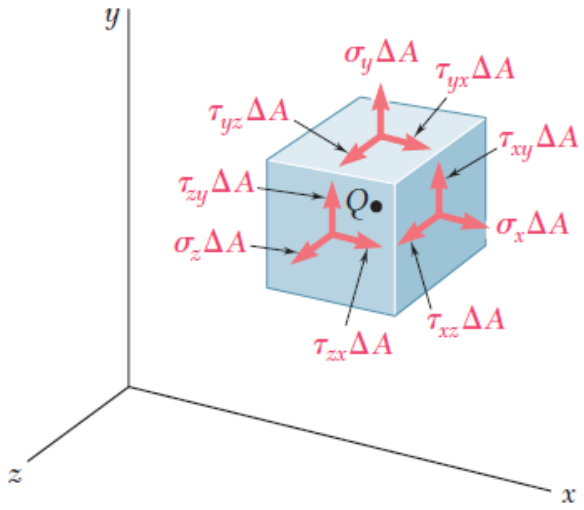


- Consider the forces in the x axis:

$$\sum F_x = 0 = \sigma_x \Delta A - \sigma'_x \Delta A$$
$$\sigma_x = \sigma'_x$$

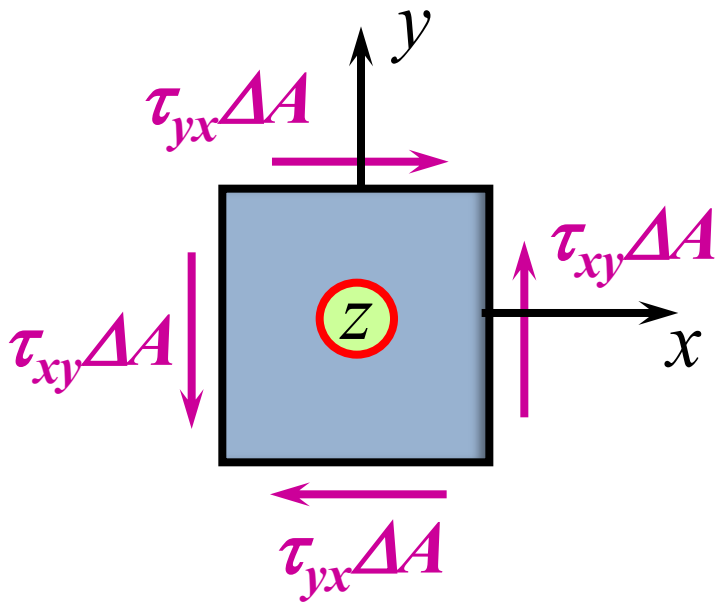
Similarly, $\sigma_y = \sigma'_y$ and $\sigma_z = \sigma'_z$

Symmetry of Stress Components



- The combination of forces generated by the stresses must satisfy the equilibrium conditions:

$$\sum F_x = \sum F_y = \sum F_z = 0$$
$$\sum M_x = \sum M_y = \sum M_z = 0$$



- Consider the moments about the *z* axis:

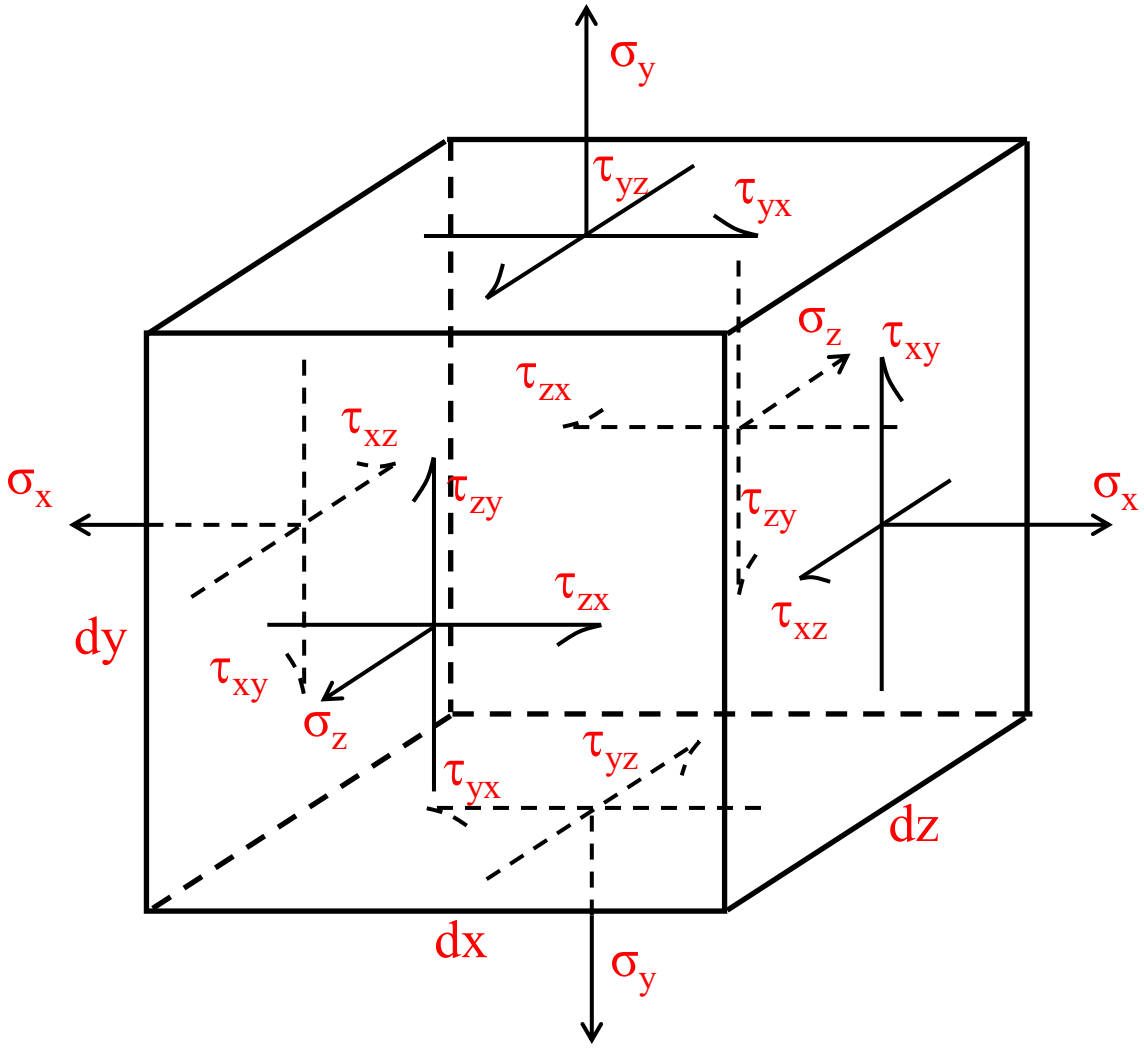
$$\sum M_z = 0 = (\tau_{xy}\Delta A)a - (\tau_{yx}\Delta A)a$$

$$\tau_{xy} = \tau_{yx}$$

Similarly, $\tau_{yz} = \tau_{zy}$ and $\tau_{xz} = \tau_{zx}$

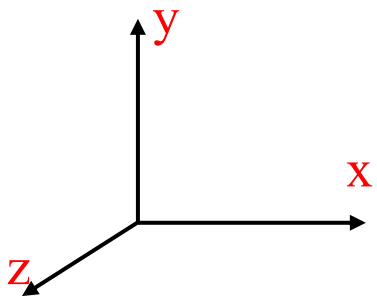
Complimentary property of shear

General Stress State of a Point



von Karman's Notation:
Symmetric

$$\left[\sigma_{ij}\right] = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$



All stress components shown in the above figure are positive.

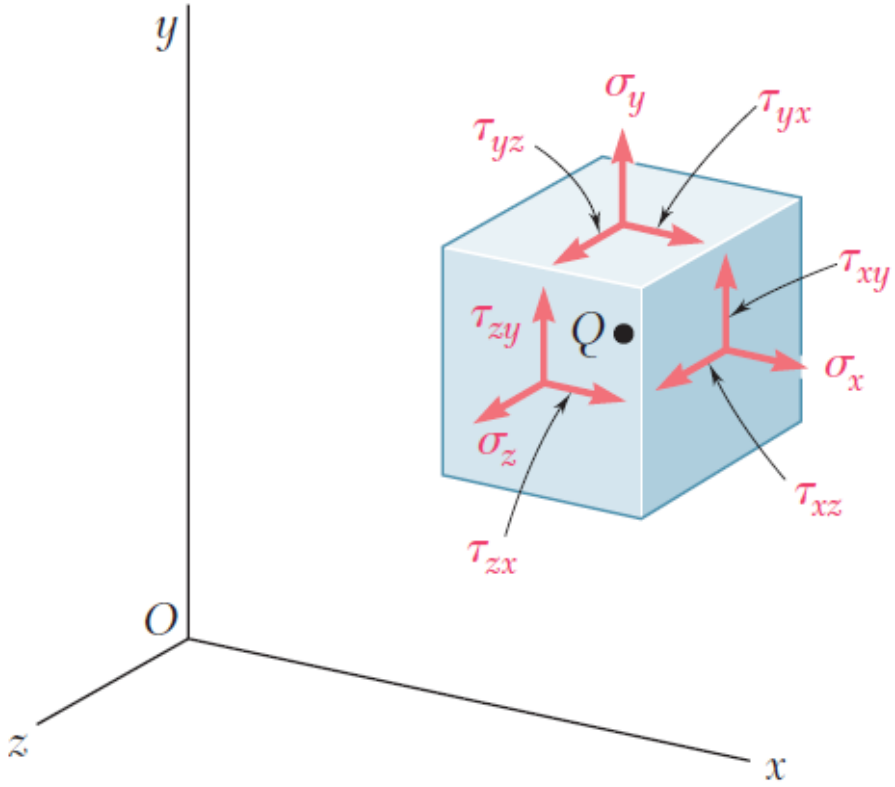
General Stress State of a Point

- The general state of stress at a point Q may be represented by 6 independent components

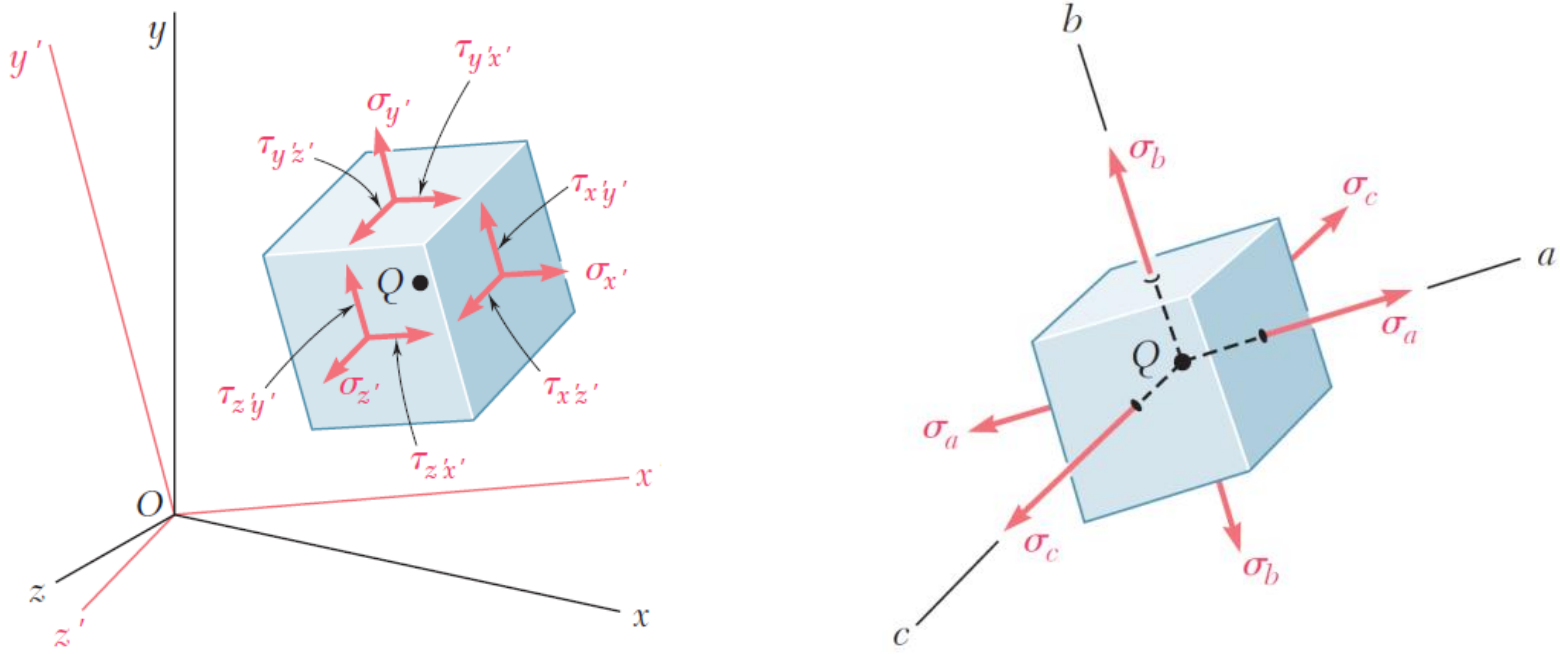
  Normal Stress : $\sigma_x, \sigma_y, \sigma_z$

  Shear Stress : $\tau_{xy}, \tau_{yz}, \tau_{zx}$

  (Note: $\tau_{xy} = \tau_{yx}, \tau_{yz} = \tau_{zy}, \tau_{zx} = \tau_{xz}$)

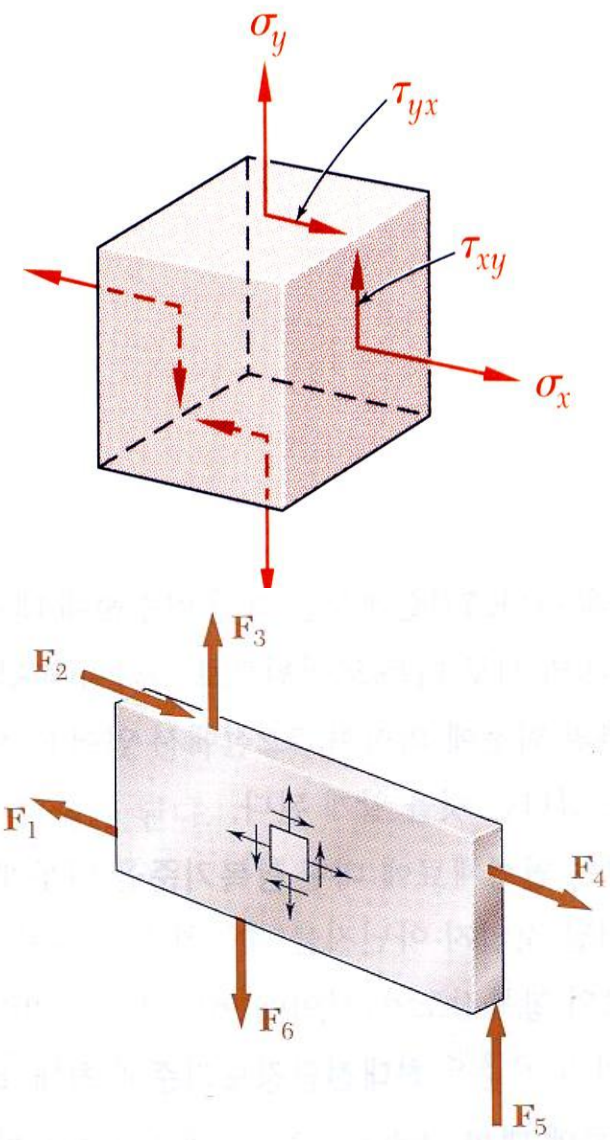


Principal Stress State of a Point



- Same stress state is represented by a different set of components when axes are rotated.
- There must exist a unique orientation (Principal Directions) of the cube, having only normal stresses (Principal Stresses) acting on each of its six faces.

Plane Stress States



- *Plane Stress* - state of stress in which two faces of the cubic element are free of stress. The state of stress is defined by

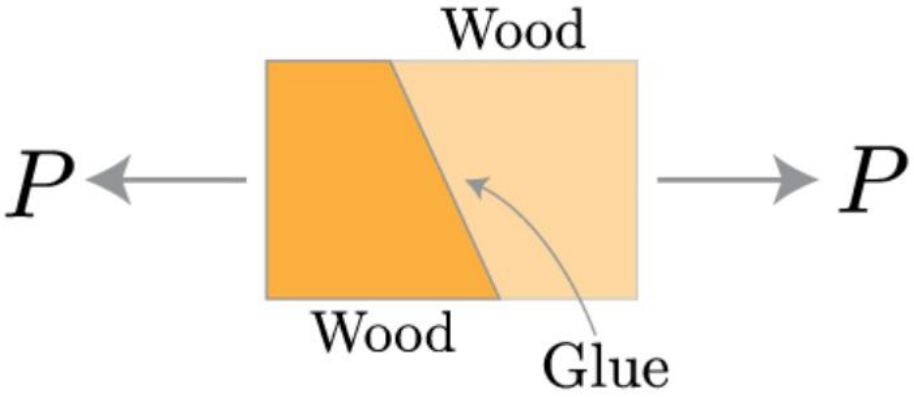
$$\sigma_x,\ \sigma_y,\ \tau_{xy} \quad \text{and} \quad \sigma_z = \tau_{zx} = \tau_{zy} = 0.$$

$$\left[\sigma_{ij}\right] = \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{yx} & \sigma_y \end{bmatrix}$$

- Example of plane stress problem:

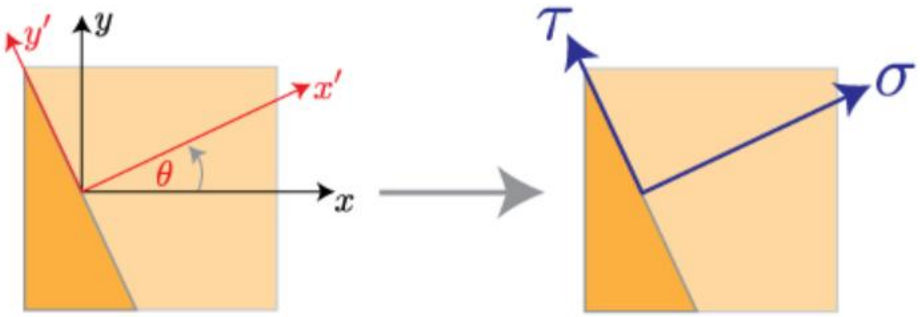
  a thin plate subjected to forces acting in the mid-plane of the plate.

Plane Stress Transformation



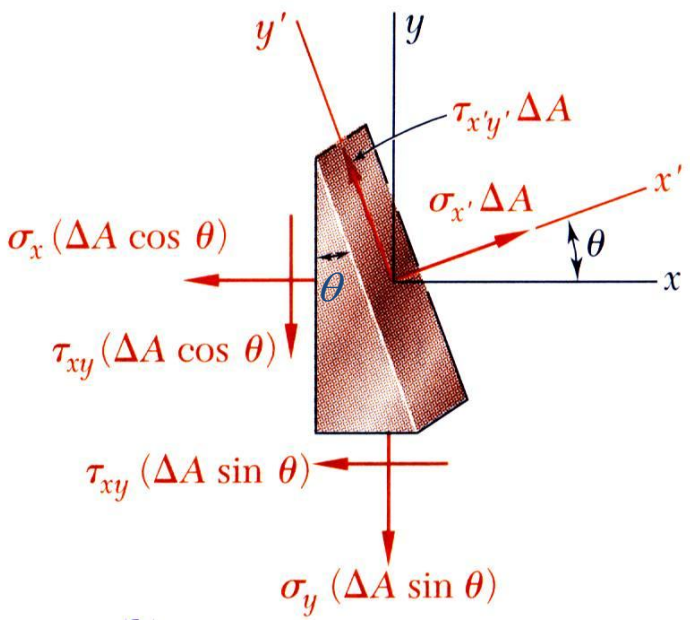
It is easy to calculate the stress at x plane.

Whether the glue can sustain the normal or shear stress?



To answer this question, we need to analyze the normal and shear stress in x' plane.

Plane Stress Transformation



- Consider force equilibrium of a prismatic element:
- Consider inclined surface with normal direction at inclined angle θ (note: counter-clockwise)

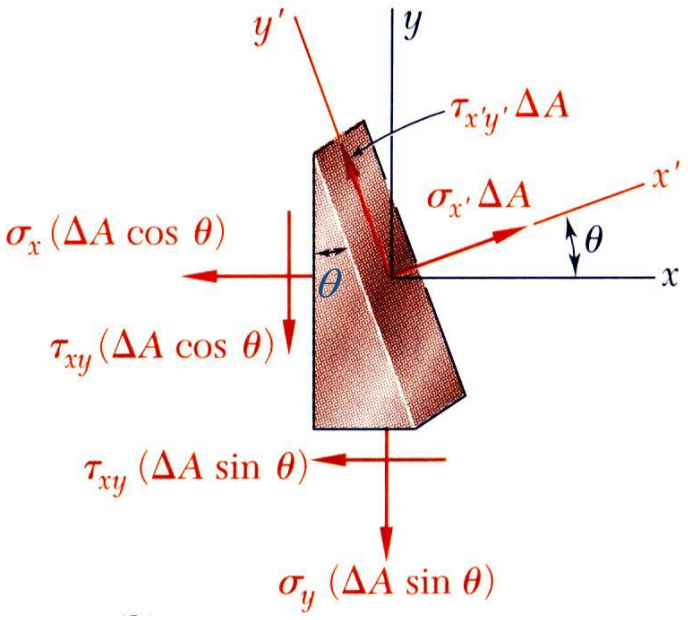
$$\sum F_{x'} = 0$$
$$= \sigma_{x'}\Delta A - \sigma_x\left(\Delta A\cos\theta\right)\cos\theta - \tau_{xy}\left(\Delta A\cos\theta\right)\sin\theta$$
$$- \sigma_y\left(\Delta A\sin\theta\right)\sin\theta - \tau_{xy}\left(\Delta A\sin\theta\right)\cos\theta$$

$$\sum F_{y'} = 0$$
$$= \tau_{x'y'}\Delta A + \sigma_x\left(\Delta A\cos\theta\right)\sin\theta - \tau_{xy}\left(\Delta A\cos\theta\right)\cos\theta$$
$$- \sigma_y\left(\Delta A\sin\theta\right)\cos\theta + \tau_{xy}\left(\Delta A\sin\theta\right)\sin\theta$$

Trigonometric Identities

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$
$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$

Plane Stress Transformation



- **Transformation equations for plane stress**

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

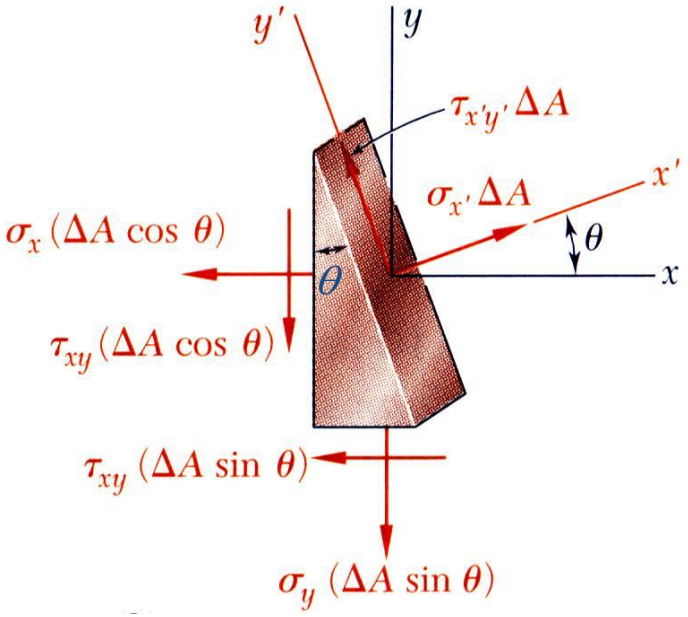
$$\left(\theta \to \theta + \pi/2\right)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2}\cos 2\theta - \tau_{xy}\sin 2\theta$$

- Observations: $\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y$

$$\frac{d\sigma_{x'}}{d\theta} = -\left(\sigma_x - \sigma_y\right)\sin 2\theta + 2\tau_{xy}\cos 2\theta = 2\tau_{x'y'}$$

Plane Stress Transformation



- **Transformation equations for plane stress**

$$\sigma_{x'} = \sigma_x \cos^2\theta + \sigma_y \sin^2\theta + 2\tau_{xy}\sin\theta\cos\theta$$

$$\sigma_{y'} = \sigma_x \sin^2\theta + \sigma_y \cos^2\theta - 2\tau_{xy}\sin\theta\cos\theta$$

$$\tau_{x'y'} = \left(\sigma_y - \sigma_x\right)\sin\theta\cos\theta + \tau_{xy}\left(\cos^2\theta - \sin^2\theta\right)$$

$$\begin{Bmatrix} \sigma_{x'} \\ \sigma_{y'} \\ \tau_{x'y'} \end{Bmatrix} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix}$$

- Matrix form: c=cosθ, s=sinθ.

Plane Stress Transformation and Strain Transformation

- **Stress Transformation matrix A**

$$\boldsymbol{\sigma}' \equiv \begin{Bmatrix} \sigma_{x'} \\ \sigma_{y'} \\ \tau_{x'y'} \end{Bmatrix} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} \equiv \mathbf{A}\boldsymbol{\sigma}$$

- For strain transformation, we also use the same **A** matrix, note engineering shear strain needs to be multiplied by 0.5 in following equation to obtain scientific shear strain.

$$\begin{Bmatrix} \epsilon_{x'} \\ \epsilon_{y'} \\ \frac{1}{2}\gamma_{x'y'} \end{Bmatrix} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{1}{2}\gamma_{xy} \end{Bmatrix} \equiv \mathbf{A} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{1}{2}\gamma_{xy} \end{Bmatrix}$$

- c=cosθ, s=sinθ. γ_{xy} is engineering shear strain.

Plane Stress Transformation

- Because **A** is orthogonal matrix: $\mathbf{A}\mathbf{A}^{\mathbf{T}}=\mathbf{A}^{\mathbf{T}}\mathbf{A}=\mathbf{I}$, the following derivation can be done to calculate stress in *xy* plane from the stress in *x'y'* plane.

$$\boldsymbol{\sigma}' = \mathbf{A}\boldsymbol{\sigma}$$

$$\mathbf{A}^{\mathbf{T}}\mathbf{A} = \mathbf{I}$$

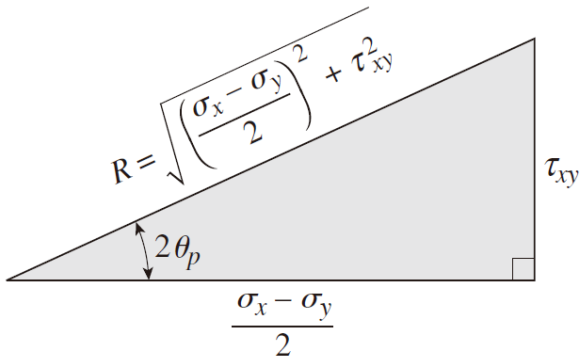
$$\mathrm{A}^{\mathrm{T}}\boldsymbol{\sigma}' = \mathrm{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{\sigma} = \boldsymbol{\sigma}$$

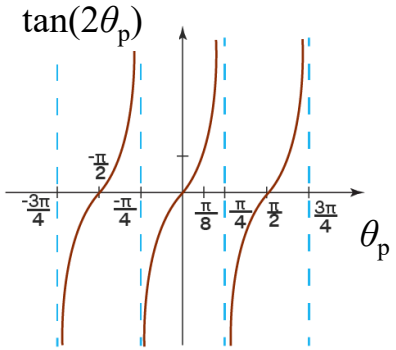
$$\boldsymbol{\sigma} \equiv \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \mathbf{A}^{\mathbf{T}}\boldsymbol{\sigma}' = \begin{bmatrix} c^2 & s^2 & -sc \\ s^2 & c^2 & sc \\ 2sc & -2sc & c^2 - s^2 \end{bmatrix} \begin{Bmatrix} \sigma_{x'} \\ \sigma_{y'} \\ \tau_{x'y'} \end{Bmatrix}$$

- $c=\cos\theta$, $s=\sin\theta$.

2D Principal Stresses

Principal stresses: the maximum and minimum value of normal stress at a point.





$$\frac{d\sigma_{x'}}{d\theta} = -\left(\sigma_x - \sigma_y\right)\sin 2\theta + 2\tau_{xy}\cos 2\theta = 2\tau_{x'y'} = 0$$

$$0 = \tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\Rightarrow \begin{cases} \sigma_{1,2} = \dfrac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ \tan 2\theta_p = \dfrac{\tau_{xy}}{\left(\sigma_x - \sigma_y\right)/2} \end{cases}$$

Note: θ_p defines two angles with a difference of 90°.

We typically limit θ_p ranging from 0 to 180 degrees. If your calculator gives you a negative value based on previous equation, then (θ_p +90°) is also the principal direction.

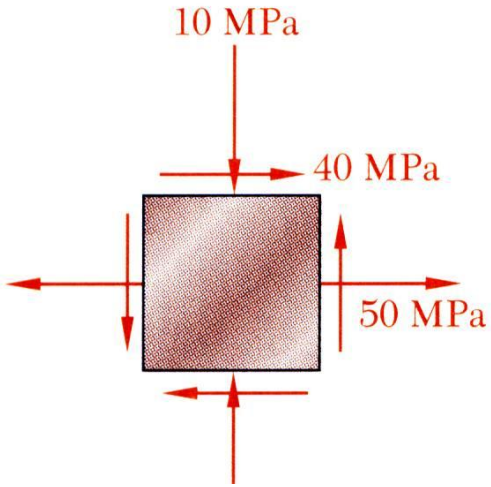
Example 7.1

For the below state of plane stress, determine

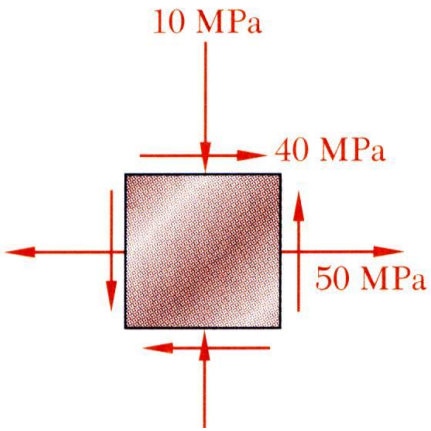
(a) the principal directions;

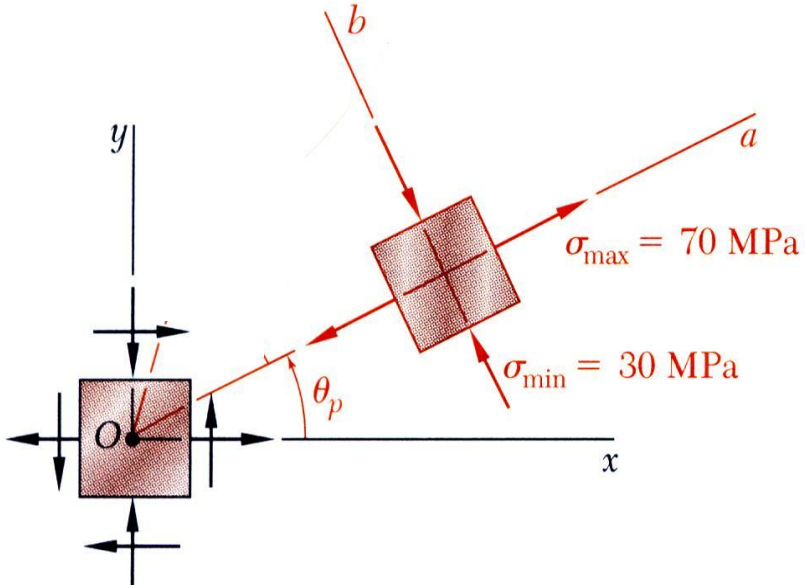
(b) the principal stresses;

and draw the state of principal plane stress.



Example 7.1





Solution:

- For current stress state,

$$\sigma_x = 50\,\text{MPa},\ \sigma_y = -10\,\text{MPa},\ \tau_{xy} = 40\,\text{MPa}$$

- Principal directions and stresses

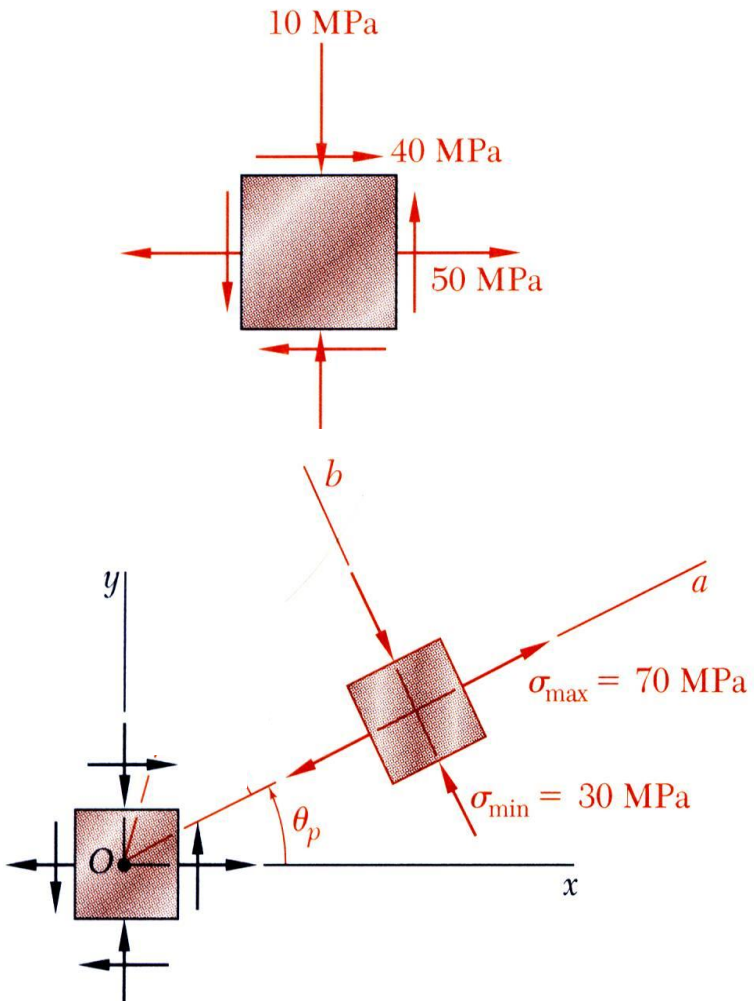
$$\sigma_{\max,\min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{50-10}{2} \pm \sqrt{\left(\frac{50+10}{2}\right)^2 + 40^2}$$

$$\sigma_{\max} = 70\,\text{MPa}$$

$$\sigma_{\min} = -30\,\text{MPa}$$

Example 7.1



Solution:

- For current stress state,

$$\sigma_x = 50\,\text{MPa},\ \sigma_y = -10\,\text{MPa},\ \tau_{xy} = 40\,\text{MPa}$$

$$\tan 2\theta_p = \frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2}$$

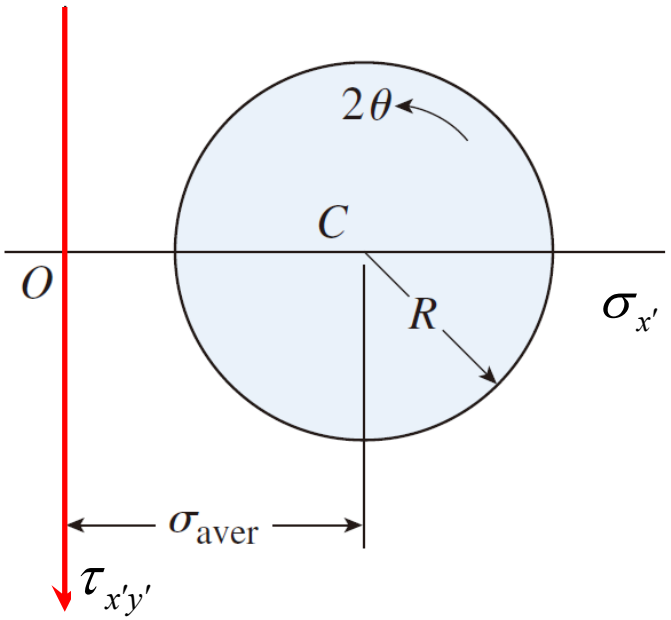
$$= \frac{40}{(50+10)/2} = \frac{4}{3}$$

$$2\theta_p = 53.1^\circ,\ 233.1^\circ$$

$$\boxed{\theta_p = 26.6^\circ,\ 116.6^\circ}$$

- Rotate counterclockwise by 26.6 degree, get the maximum stress of 70 MPa.
- Need to use stress transformation equation to find which angle is maximum and which is minimum.

Mohr's Circle



$$\left.\begin{aligned}\sigma_{x'} - \frac{\sigma_x + \sigma_y}{2} &= \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta \\ \tau_{x'y'} &= -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta\end{aligned}\right\} \Rightarrow$$

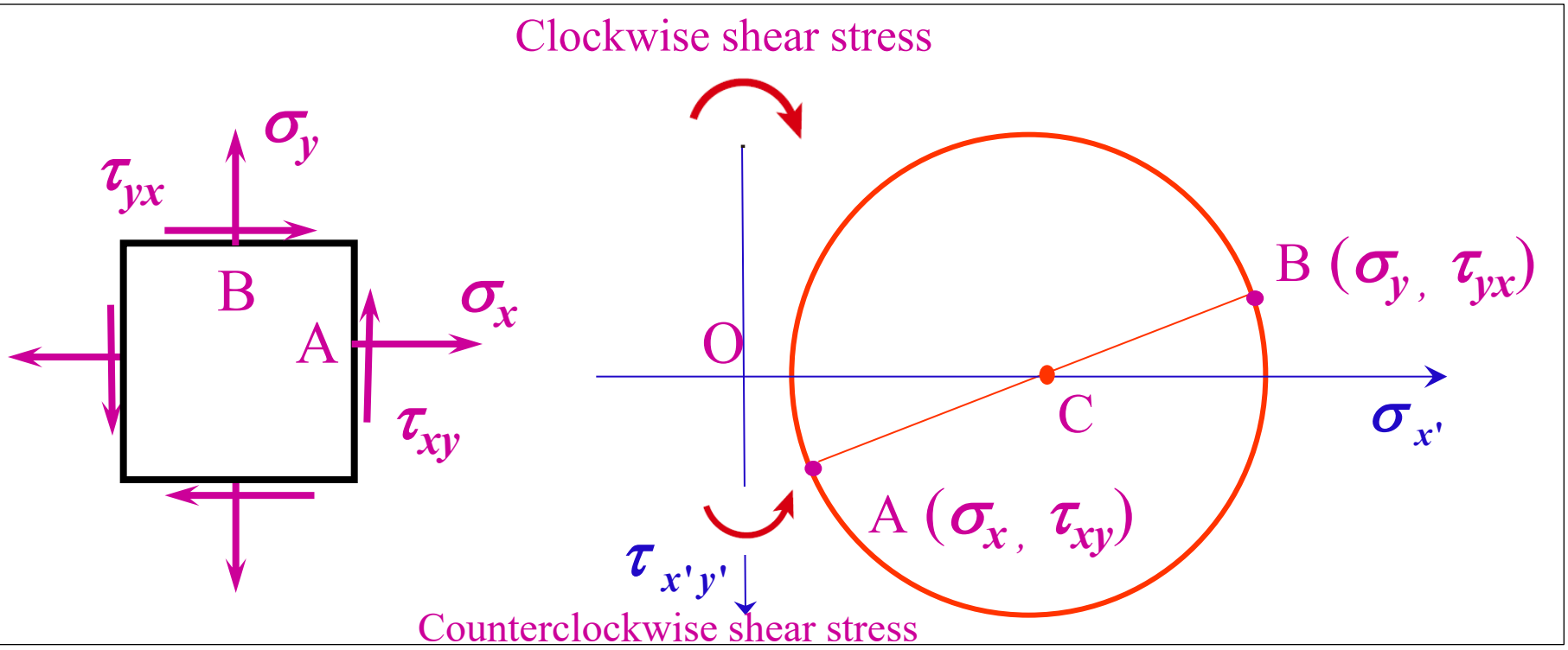
$$\left(\sigma_{x'} - \sigma_{ave}\right)^2 + \tau_{x'y'}^2 = R^2$$

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

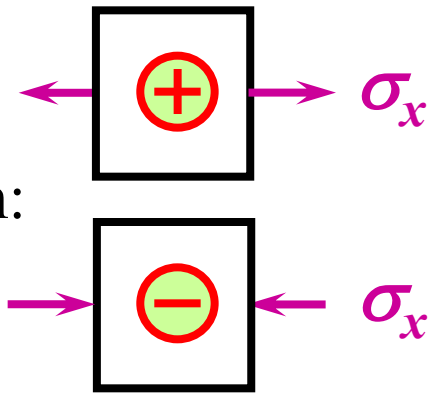
$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

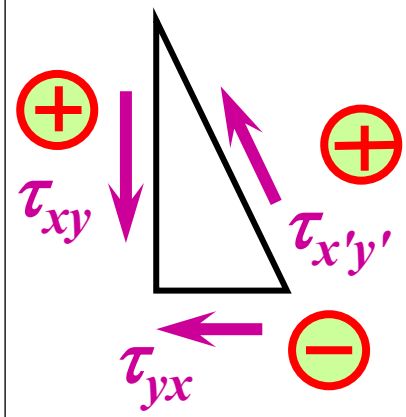
- Note: in our definition, in Mohr's circle, shear stress $\tau_{x'y'}$ is positive downward and the angle 2θ is positive counterclockwise. Please remember this sign convention. Some other books have other definition.
- The advantage of plotting shear stresses positive downward is that the angle 2θ on Mohr's circle will be positive when counterclockwise, which agrees with the positive direction of 2θ in the derivation of the transformation equations

Construct Mohr's Circle



Sign:





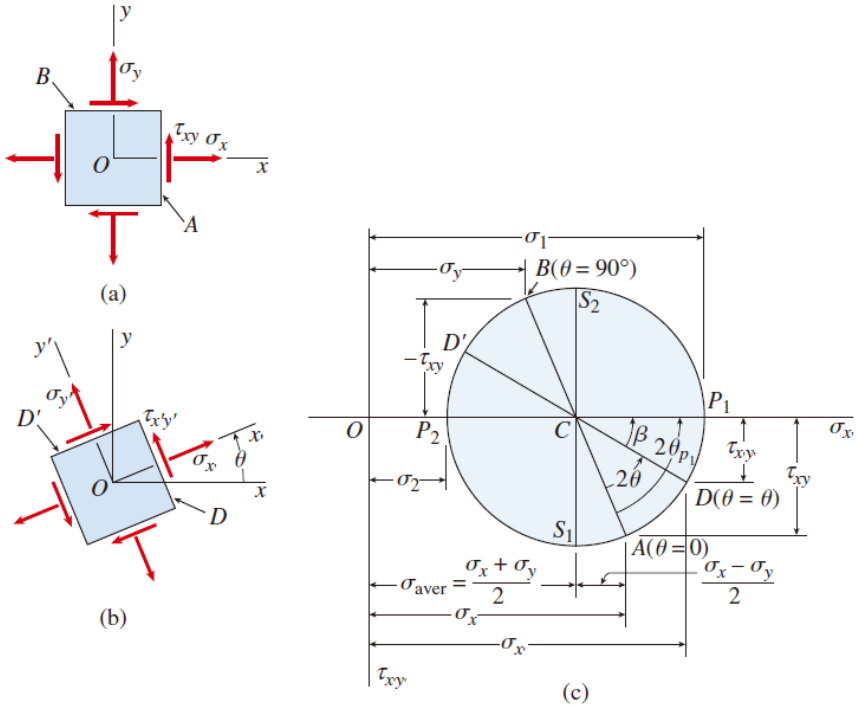
Counterclockwise shear stress is positive, plotted downward In Mohr circle.

$$(\sigma_{x'} - \sigma_{ave})^2 + \tau_{x'y'}^2 = R^2$$

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Constructing Mohr's Circle



$$\left(\sigma_{x'} - \sigma_{ave}\right)^2 + \tau_{x'y'}^2 = R^2$$

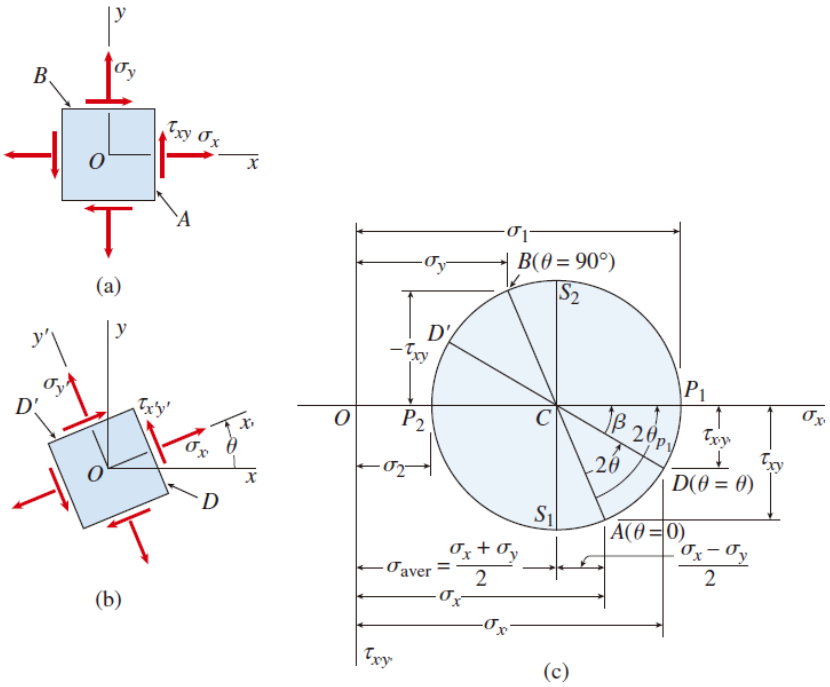
$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

With σ_x, σ_y, and τ_{xy} known, the **procedure for constructing Mohr's circle** is as follows (Fig. c):

1. Draw a set of coordinate axes with $\sigma_{x'}$ as abscissa (positive to the right) and $\tau_{x'y'}$ as ordinate (positive downward).
2. Locate the center C of the circle at the point having coordinates $\sigma_{ave}=0.5(\sigma_x+\sigma_y)$ and shear stress 0.
3. Locate point A, representing the stress conditions on the x face of the element shown in Fig. a, by plotting its coordinates $\sigma_{x'}= \sigma_x$ and shear stress $\tau_{x'y'}=\tau_{xy}$. Note that point A on the circle corresponds to angle $\theta=0$. Also, note that the x face of the element (Fig. a) is labeled "A" to show its correspondence with point A on the circle.
4. Locate point B, representing the stress conditions on the y face of the element shown in Fig. a, by plotting its coordinates $\sigma_{x'}= \sigma_y$ and $\tau_{x'y'}=-\tau_{xy}$. Note that point B on the circle corresponds to $\theta=90^o$. The y face of element is labeled B and marked on circle.
5. Using point C as center, draw Mohr's circle through points A and B. The circle will have radius R.

Plane Stress Transformation in Mohr's Circle



On the circle, measure an angle 2θ counterclockwise from radius CA, because point A corresponds to $\theta=0$ and is the reference point from which we measure angles. The angle 2θ locates point D on the circle. It can be proven that the stress state corresponding to point D on the Mohr's circle is as follows:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$$

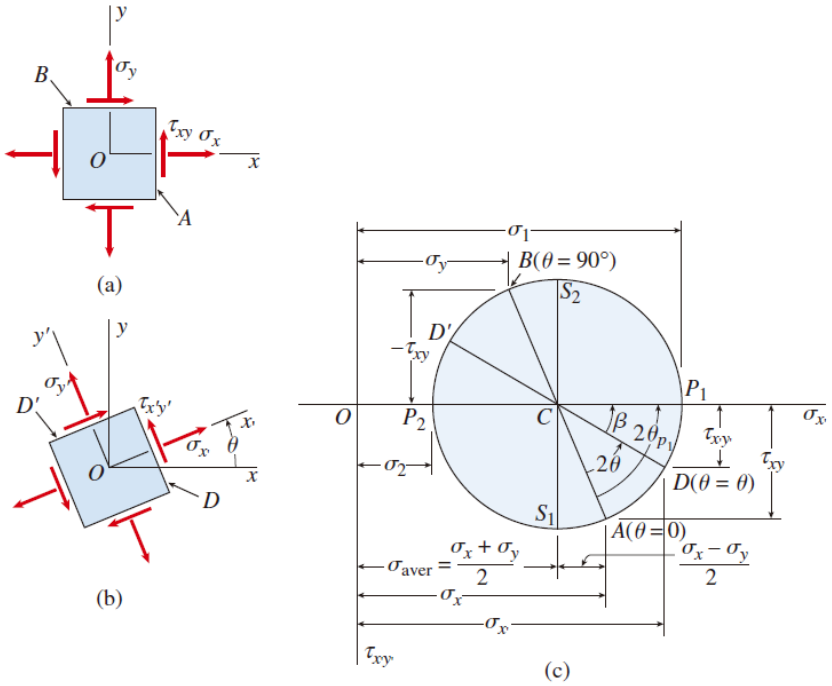
$$\left(\sigma_{x'} - \sigma_{ave}\right)^2 + \tau_{x'y'}^2 = R^2$$

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Conclusion: as we rotate the axes counterclockwise through an angle θ (Fig. b), the point on Mohr's circle corresponding to the x' face moves counterclockwise through an angle 2θ.

Principal stress Mohr's Circle



$$\left(\sigma_{x'} - \sigma_{ave}\right)^2 + \tau_{x'y'}^2 = R^2$$

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

At point P_1 in Mohr's circle, we encounter the maximum normal stress and shear stress is zero. P_1 represents a principal stress and a principal plane.

$$\sigma_1 = OC + \overline{CP_1} = \frac{\sigma_x + \sigma_y}{2} + R$$

$$\cos 2\theta_{p_1} = \frac{\sigma_x - \sigma_y}{2R} \qquad \sin 2\theta_{p_1} = \frac{\tau_{xy}}{R}$$

$$\tan 2\theta_p = \frac{\tau_{xy}}{\left(\sigma_x - \sigma_y\right)/2}$$

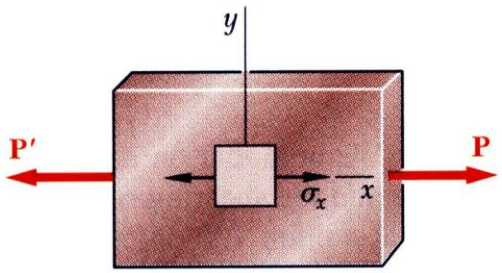
At point P_2 in Mohr's circle, we encounter the minimum normal stress and shear stress is zero.

$$\theta_{p_2} = \theta_{p_1} + 90°$$

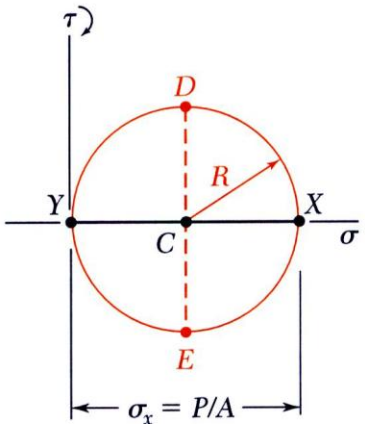
At points S_1 and S_2, shear stress are maximum or minimum and normal stress equals average stress σ_{ave}.

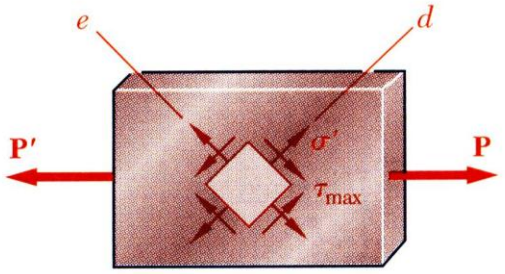
Example 7.2

- Bar under uniaxial loading:



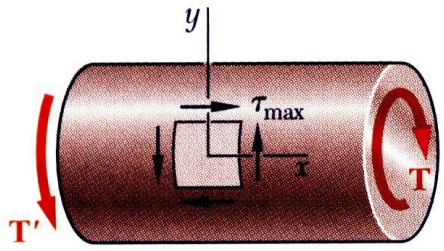
$$\sigma_x = \frac{P}{A}, \quad \sigma_y = \tau_{xy} = 0$$



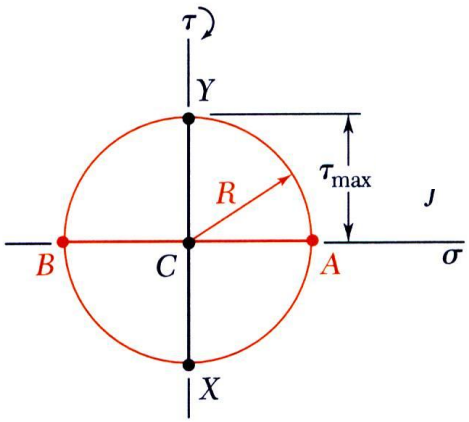


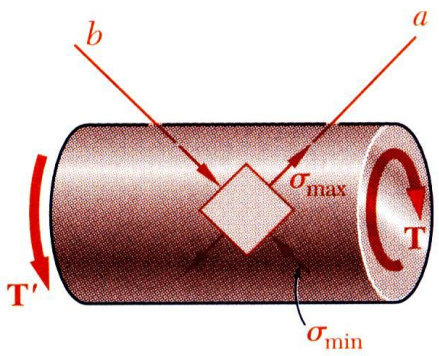
$$\sigma_x = \sigma_y = \tau_{xy} = \frac{P}{2A}$$

- Circular shaft under torque:



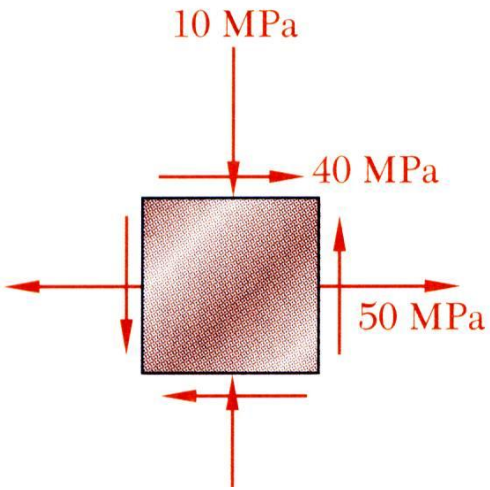
$$\sigma_x = \sigma_y = 0 \quad \tau_{xy} = \frac{T}{W_P}$$





$$\sigma_x = \frac{T}{W_P} \quad \sigma_y = -\frac{T}{W_P} \quad \tau_{xy} = 0$$

Example 7.3



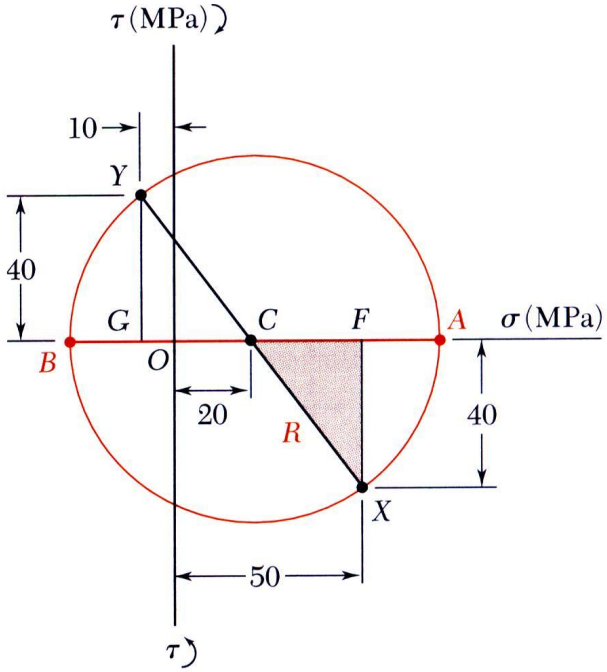
For the above state of plane stress construct Mohr's circle to determine

(a) the principal planes,

(b) the principal stresses,

(c) the maximum shear stress

and the corresponding normal stress.



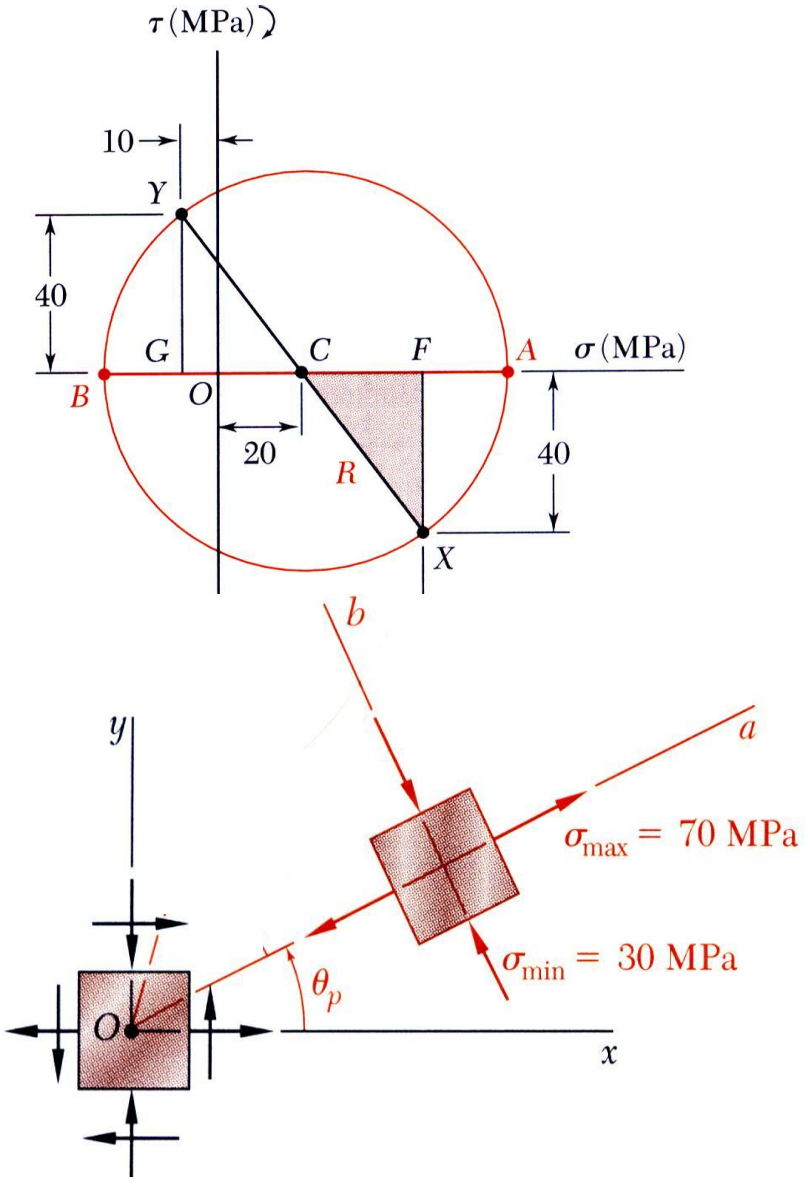
Solution:

- Construction of Mohr's circle

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = \frac{(50)+(-10)}{2} = 20\,\text{MPa}$$

$$CF = 50 - 20 = 30\,\text{MPa} \qquad FX = 40\,\text{MPa}$$

$$R = CX = \sqrt{(30)^2 + (40)^2} = 50\,\text{MPa}$$



- Principal planes and stresses

$$\sigma_{\max} = OA = OC + CA = 20 + 50$$

$$\boxed{\sigma_{\max} = 70\,\text{MPa}}$$

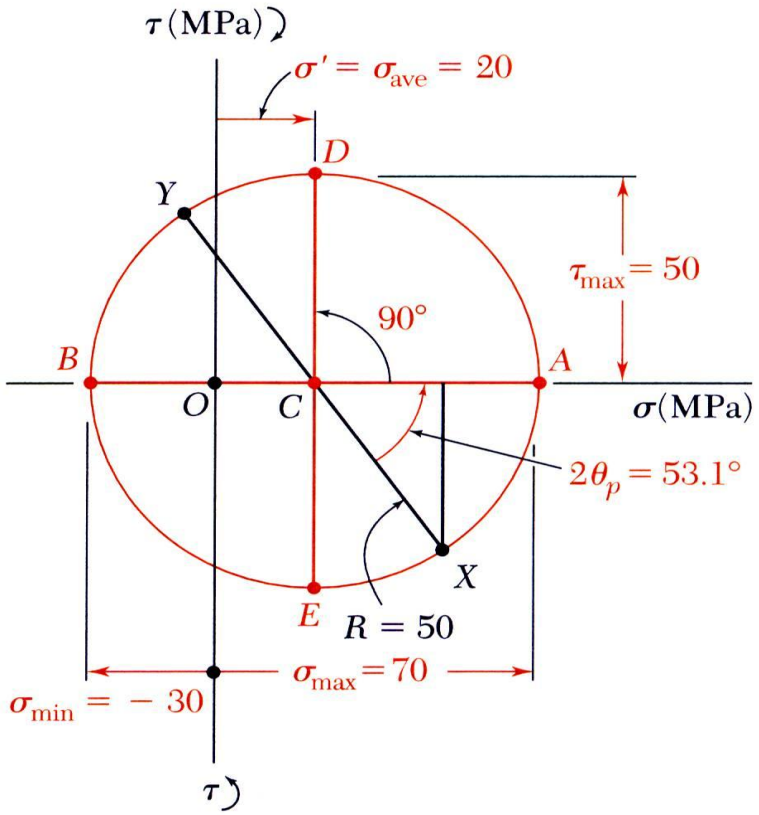
$$\sigma_{\min} = OB = OC - BC = 20 - 50$$

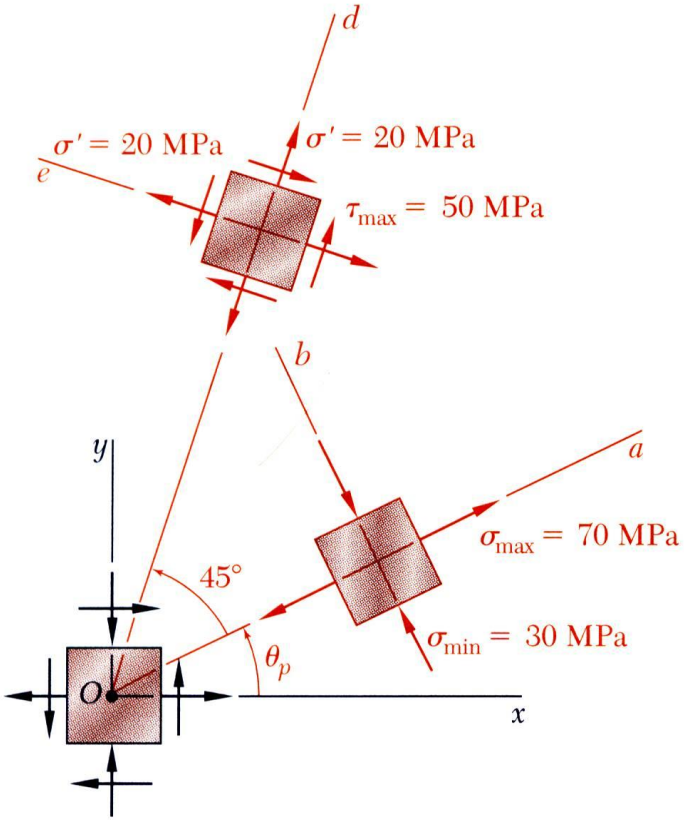
$$\boxed{\sigma_{\min} = -30\,\text{MPa}}$$

$$\tan 2\theta_p = \frac{FX}{CF} = \frac{40}{30}$$

$$2\theta_p = 53.1°,\ 233.1°$$

$$\boxed{\theta_p = 26.6°,\ 116.6°}$$





- Maximum shear stress

$\theta_s = \theta_p + 45°$ $\quad$ $\tau_{\max} = R$ $\quad$ $\sigma' = \sigma_{ave}$

$\theta_s = 71.6°,\ 161.6°$ $\quad$ $\tau_{\max} = 50\,\text{MPa}$ $\quad$ $\sigma' = 20\,\text{MPa}$

Ordering of Principal Stress State

- Stress states classified based on the number of zero principal stresses:

- Tri-axial stress state:

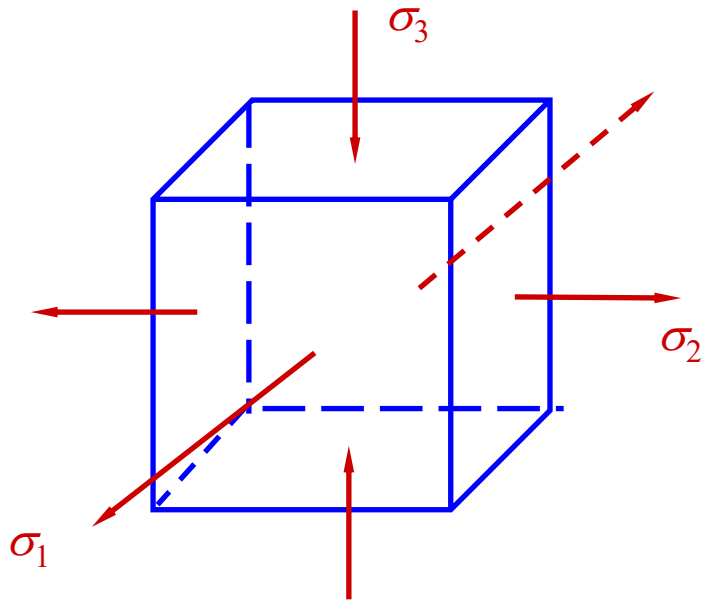
  none zero principal stresses;

- Biaxial stress state:

  one zero principal stresses;

- Uniaxial stress state:

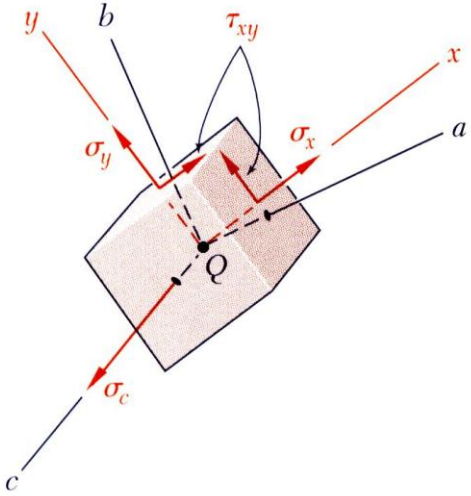
  two zero principal stresses;

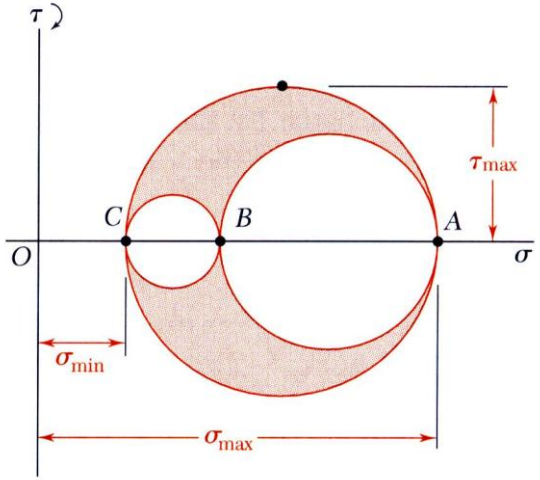


Principal directions & stresses

$$\sigma_1 \geq \sigma_2 \geq \sigma_3$$

Application of Mohr's Circle in 3-D





- Points *A*, *B*, and *C* represent the principal stresses on the principal planes (shear stress is zero)
- The three circles represent the normal and shear stresses for rotation around each principal axis.
- Radius of the largest circle yields the maximum shear stress.

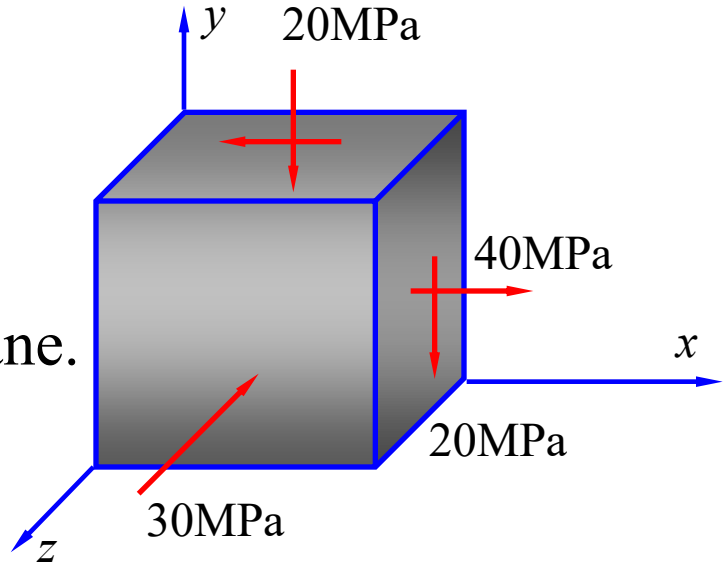
$$\tau_{max} = \frac{1}{2}|\sigma_{max} - \sigma_{min}|$$

Example 7.4

- For the state of 3-D stress shown below, determine the principal stresses and maximum shear stress.
- Solution

1. Summarize the 3-D stress state.

$$[\sigma]=\begin{bmatrix} 40 & -20 & 0 \\ -20 & -20 & 0 \\ 0 & 0 & -30 \end{bmatrix}$$



2. Find the two principal stresses in x-y plane.

$$\left.\begin{matrix}\sigma_i \\ \sigma_j\end{matrix}\right\} = \frac{40-20}{2} \pm \sqrt{\left(\frac{40+20}{2}\right)^2 + 20^2} = \begin{cases} 46 \text{ MPa} \\ -26 \text{ MPa} \end{cases}$$

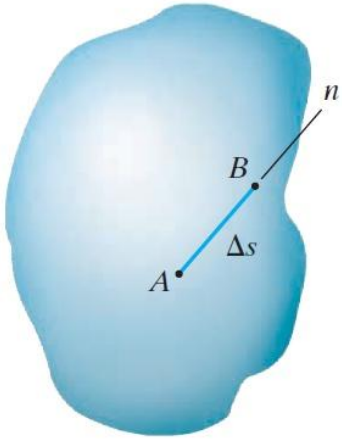
3. Rearrange the three principal stresses.

$$\sigma_1 = 46 \text{ MPa}, \sigma_2 = -26 \text{ MPa}, \sigma_3 = -30 \text{ MPa}$$

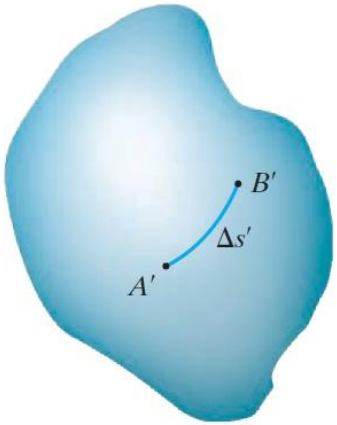
4. Find the maximum shear stress.

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{46-(-30)}{2} = 38 \text{ MPa}$$

Normal Strain and Shear Strain



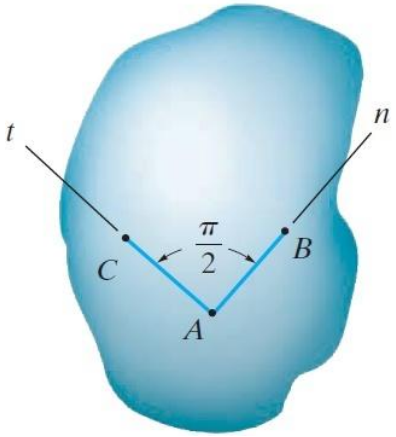
Undeformed



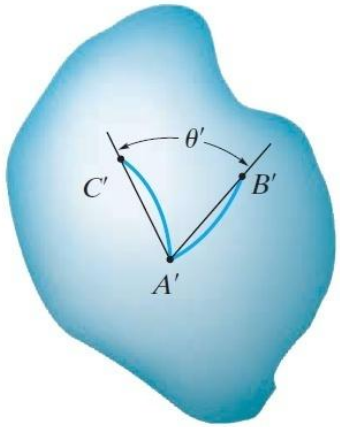
Deformed

Normal Strain

$$\varepsilon = \lim_{\text{B}\to\text{A along } n} \frac{\Delta s' - \Delta s}{\Delta s}$$



Undeformed

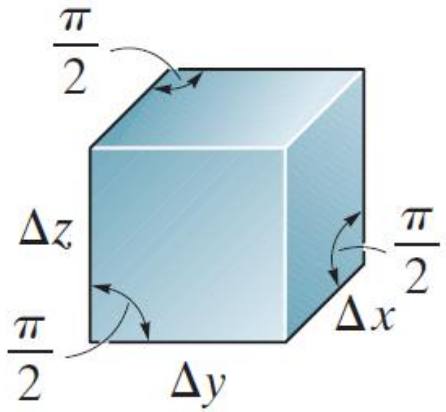


Deformed

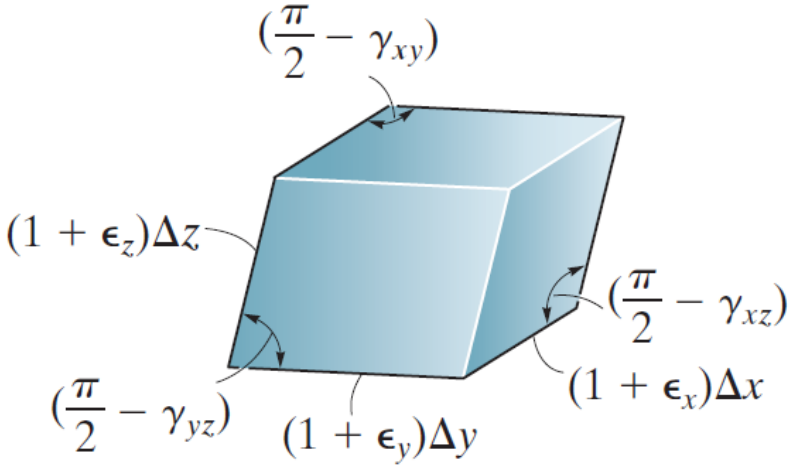
Shear Strain

$$\gamma = \frac{\pi}{2} - \lim_{\substack{\text{B}\to\text{A along } n\,,\\ \text{C}\to\text{A along } t}} \theta'$$

Strain State



Undeformed



Deformed

- Strain state can be represented as

$$\left[\varepsilon_{ij}\right]=\begin{bmatrix}\varepsilon_x & \gamma_{xy} & \gamma_{xz}\\ \gamma_{yx} & \varepsilon_y & \gamma_{yz}\\ \gamma_{zx} & \gamma_{zy} & \varepsilon_z\end{bmatrix}$$

Normal Strain : $\varepsilon_x, \varepsilon_y, \varepsilon_z$

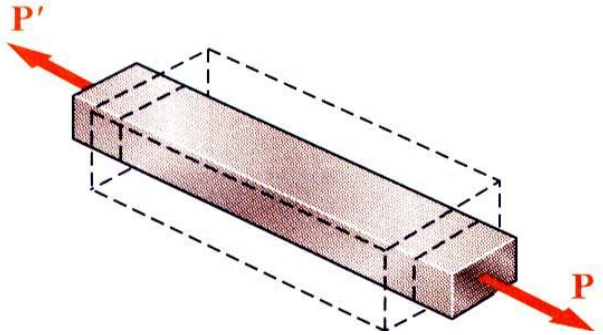
Shear Strain : $\gamma_{xy}, \gamma_{yz}, \gamma_{zx}$

(Note: $\gamma_{xy}=\gamma_{yx}, \gamma_{yz}=\gamma_{zy}, \gamma_{zx}=\gamma_{xz}$)

- Normal strains cause a change in volume, whereas shear strains cause a change in its shape;
- Question: Volume change ratio in terms of strain state?

Generalized Hooke's Law

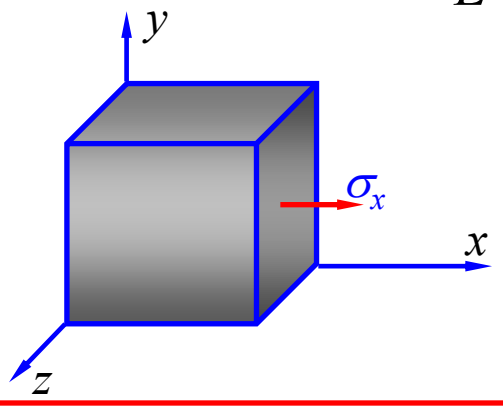
- For a bar under uniaxial tension:
- Axial strain: $\varepsilon = \sigma / E$
- Transverse strain: $\varepsilon' = -\nu\varepsilon = -\nu\sigma / E$



Uniaxial stress along x:

$$\varepsilon_x = \frac{\sigma_x}{E},$$

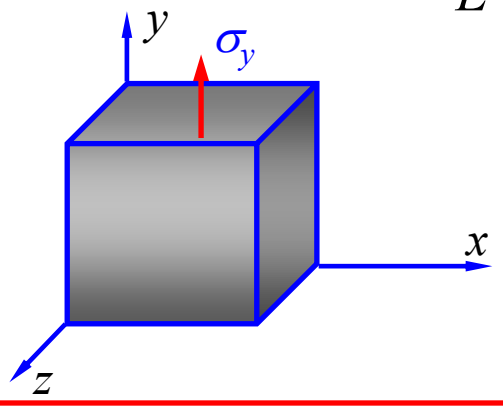
$$\varepsilon_y = \varepsilon_z = -\nu\varepsilon_x = -\nu\frac{\sigma_x}{E}$$



Along y:

$$\varepsilon_y = \frac{\sigma_y}{E},$$

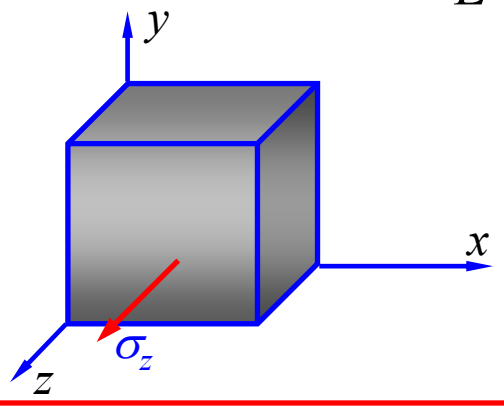
$$\varepsilon_z = \varepsilon_x = -\nu\varepsilon_y = -\nu\frac{\sigma_y}{E}$$



Along z:

$$\varepsilon_z = \frac{\sigma_z}{E},$$

$$\varepsilon_x = \varepsilon_y = -\nu\varepsilon_z = -\nu\frac{\sigma_z}{E}$$

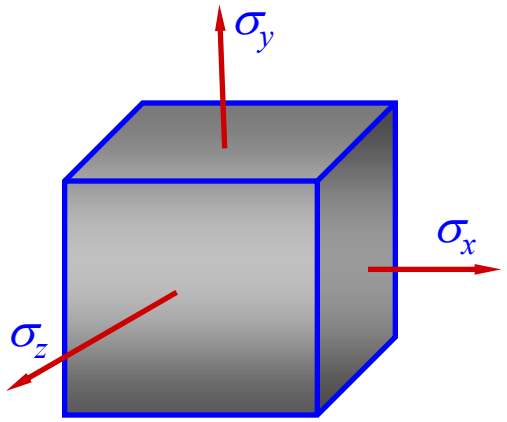


Generalized Hooke's Law (Isotropic Material)

$$\varepsilon_x = \frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} - \nu\frac{\sigma_z}{E} = \frac{(1+\nu)}{E}\sigma_x - \frac{\nu}{E}(\sigma_x + \sigma_y + \sigma_z)$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \nu\frac{\sigma_z}{E} - \nu\frac{\sigma_x}{E} = \frac{(1+\nu)}{E}\sigma_y - \frac{\nu}{E}(\sigma_x + \sigma_y + \sigma_z)$$

$$\varepsilon_z = \frac{\sigma_z}{E} - \nu\frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} = \frac{(1+\nu)}{E}\sigma_z - \frac{\nu}{E}(\sigma_x + \sigma_y + \sigma_z)$$

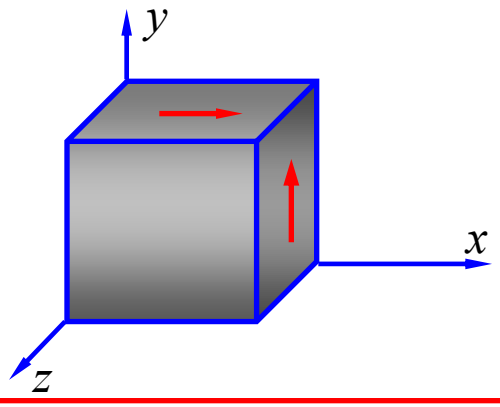


Shear Stresses

$$\boxed{E = 2G(1+\nu)}$$

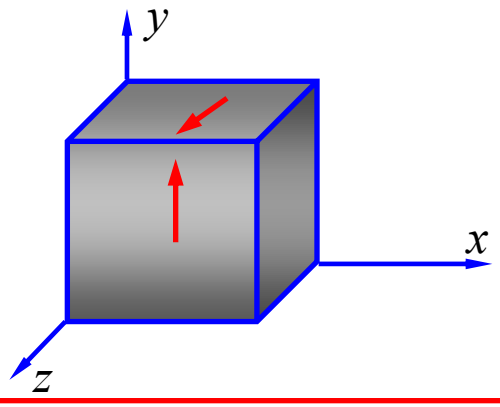
Shear stress around z:

$$\tau_{xy} = G\gamma_{xy}$$



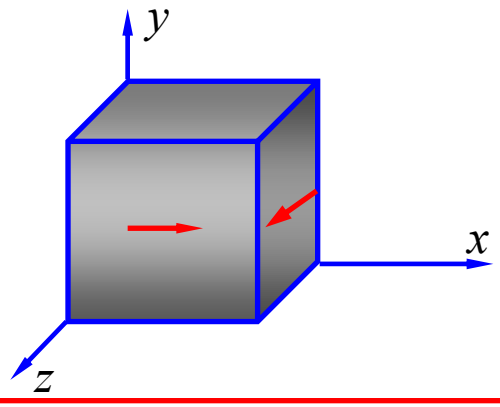
Around x:

$$\tau_{yz} = G\gamma_{yz}$$



Around y:

$$\tau_{zx} = G\gamma_{zx}$$



Generalized Hooke's Law for plane stress

Out-of-plane stress σ_z=0 compared to 3D case:

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y)$$

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x)$$

$$\varepsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y)$$

First two equations may be re-written as below:

$$\sigma_x = \frac{E}{1-\nu^2}(\varepsilon_x + \nu\varepsilon_y)$$

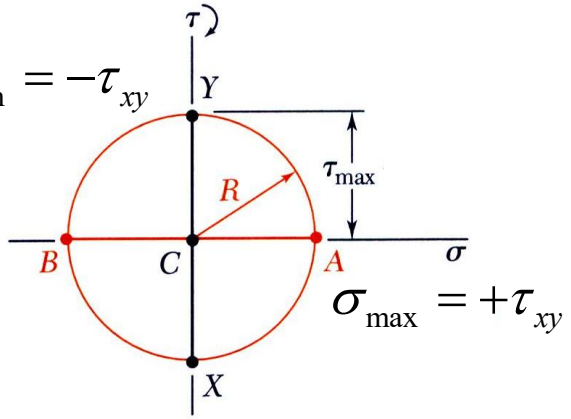
$$\sigma_y = \frac{E}{1-\nu^2}(\varepsilon_y + \nu\varepsilon_x)$$

Relationship between *E* and *G*

- Consider a small part of material under pure shear stress τ_{xy} and pure shear strain $\gamma_{xy} = \tau_{xy}/G$
- The strain energy per unit volume u in pure shear is:

$$u = \frac{1}{2}\gamma_{xy}\tau_{xy} = \frac{\tau_{xy}^2}{2G}$$

- Based on stress transformation matrix, the shear stress is equivalent to two principal stresses based on Mohr's circle:



$\sigma_{min} = -\tau_{xy}$

$\sigma_{max} = +\tau_{xy}$

$$\sigma_{max} = +\tau_{xy} \quad \sigma_{min} = -\tau_{xy}$$

$$\varepsilon_{max} = \frac{\tau_{xy}}{E} + \nu\frac{\tau_{xy}}{E}$$

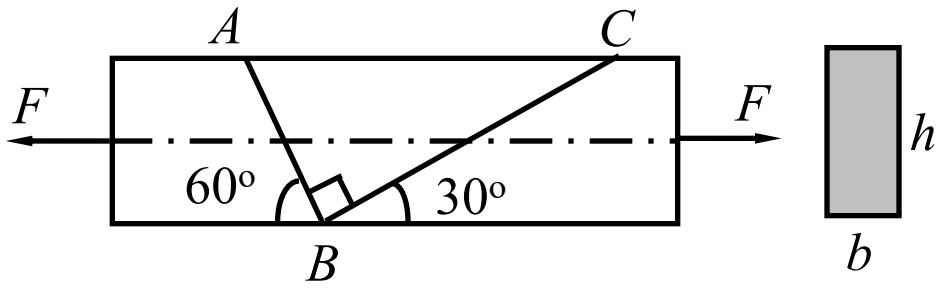
$$\varepsilon_{min} = -\frac{\tau_{xy}}{E} - \nu\frac{\tau_{xy}}{E}$$

Strain energy per unit volume: $u = 0.5\sigma_{max}\varepsilon_{max} + 0.5\sigma_{min}\varepsilon_{min} = \frac{\tau_{xy}^2}{E}(1+\nu)$

Because strain energy per unit volume is a constant (not change with respect to coordinate system): $G = \frac{E}{2(1+\nu)}$

Example 7.5 Solution method 1

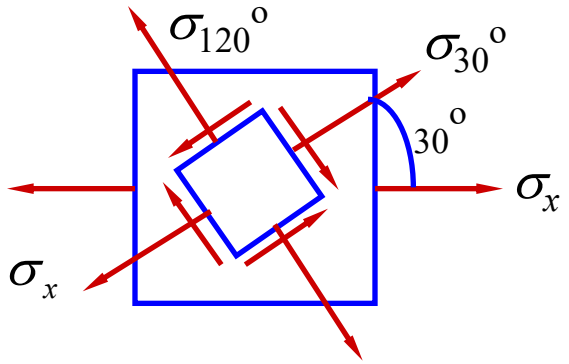
- For the bar shown, find the change in length of line segment *BC* and in right angle *ABC*. Assume the Young's modulus and Poisson's ratio as *E* and *v* respectively.
- Solution: θ=30º



1. Stress state in reference coordinate system

$$\sigma_x = F/bh$$

All other stress components vanish.



2. Stress state in transformed coordinates:

$$\sigma_{30^0} = \sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta = \tfrac{3}{4}\sigma_x$$

$$\tau_{30^0} = \tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta = -\tfrac{\sqrt{3}}{4}\sigma_x$$

$$\sigma_{120^0} = \sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2}\cos 2\theta - \tau_{xy}\sin 2\theta = \tfrac{1}{4}\sigma_x$$

Example 7.5 Solution method 1

3. Longitudinal strain along BC

$$\varepsilon_{30^0} = \frac{1}{E}\left(\sigma_{30^0} - \nu\sigma_{120^0}\right) = \frac{3-\nu}{4E}\sigma_x = \frac{(3-\nu)F}{4Ebh}$$

4. The change in length of BC

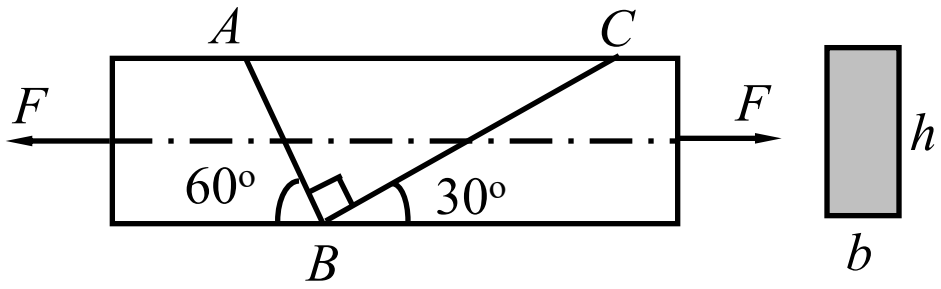
$$\Delta l_{BC} = \varepsilon_{30^0} \times l_{BC} = 2h\varepsilon_{30^0} = \frac{(3-\nu)F}{2Eb}$$

5. The change in right angle ABC

$$\gamma = \frac{\tau_{30^0}}{G} = \frac{-\frac{\sqrt{3}}{4}\sigma_x}{\frac{E}{2(1+\nu)}} = -\frac{\sqrt{3}(1+\nu)}{2E}\sigma_x = -\frac{\sqrt{3}}{2}\frac{(1+\nu)F}{bhE}$$

Example 7.5 Solution method 2

- For the bar shown, find the change in length of line segment *BC* and in right angle *ABC*. Assume the Young's modulus and Poisson's ratio as *E* and *v* respectively.
- Solution



1. Stress state in reference coordinate system

$$\sigma_x = F/bh$$

All other stress components vanish.

2. Strain state in reference coordinates:

$$\varepsilon_x = \frac{\sigma_x}{E} = \frac{F}{Ebh}$$

$$\varepsilon_y = -\frac{v\sigma_x}{E} = \frac{-vF}{Ebh}$$

$$\gamma_{xy} = 0$$

Example 7.5 Solution method 2

3. Strain transformation matrix $A(\theta=30^{o})$

$$\begin{Bmatrix} \epsilon_{x'} \\ \epsilon_{y'} \\ \frac{1}{2}\gamma_{x'y'} \end{Bmatrix} = \begin{bmatrix} c^2 & s^2 & 2sc \\ s^2 & c^2 & -2sc \\ -sc & sc & c^2 - s^2 \end{bmatrix} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{1}{2}\gamma_{xy} \end{Bmatrix} = \mathbf{A} \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{1}{2}\gamma_{xy} \end{Bmatrix} = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & \frac{\sqrt{3}}{2} \\ \frac{1}{4} & \frac{3}{4} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{4} & \frac{\sqrt{3}}{4} & \frac{1}{2} \end{bmatrix} \begin{Bmatrix} \frac{F}{Ebh} \\ \frac{-\nu F}{Ebh} \\ 0 \end{Bmatrix} = \begin{Bmatrix} \frac{(3-\nu)F}{4Ebh} \\ \frac{(1-3\nu)F}{4Ebh} \\ -\frac{\sqrt{3}(1+\nu)F}{4bh} \end{Bmatrix}$$

5. The change in right angle ABC

$$\gamma = -\frac{\sqrt{3}}{2}\frac{(1+\nu)F}{bhE}$$

Summary

- Stresses on inclined sections cut through a body may be larger than stresses acting on a stress element aligned with cross section.
- Equilibrium of a wedge element is used to derive the stress transformation relationship, applicable to stresses in any kind of material.

$$\sigma_{x'} = \sigma_x \cos^2\theta + \sigma_y \sin^2\theta + 2\tau_{xy} \sin\theta\cos\theta$$

$$\sigma_{y'} = \sigma_x \sin^2\theta + \sigma_y \cos^2\theta - 2\tau_{xy} \sin\theta\cos\theta$$

$$\tau_{x'y'} = (\sigma_y - \sigma_x)\sin\theta\cos\theta + \tau_{xy}(\cos^2\theta - \sin^2\theta)$$

- The principal stresses (maximum and minimum normal stresses) are found from the transformation equation:

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Summary

- The principal shear stress can be obtained as (equal the diameter of Mohr's circle):

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

- The transformation equations for plane stress can be represented graphically by a plot of Mohr's circle, which displays relationship between normal and shear stress acting on arbitrary inclined planes.

- Strain transformation relationship in plane strain (ε_z=0):

$$\varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\frac{\gamma_{x_1y_1}}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$